A FIELD-BASED PROJECT REPORT

Decarbonisation Pathways in Climate-Constrained Energy Systems: A Hybrid Modelling Approach

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Abstract

This study investigates decarbonisation pathways within climate-constrained energy systems in India by integrating energy, economy, and environmental dimensions through a hybrid modelling framework. Anchored in the commitments of the Paris Agreement and Sustainable Development Goals (SDGs), particularly SDG 7, SDG 8, and SDG 13, the research explores how India can balance economic growth with greenhouse gas (GHG) mitigation.

The study employs the IMACLIM-IND hybrid model, which combines top-down Computable General Equilibrium (CGE) modelling with bottom-up energy system modelling (AIM/Enduse), enabling a comprehensive analysis of structural, technological, and economic dynamics. A hybrid Input-Output (IO) table is constructed by reconciling energy balances with national accounts, incorporating sectoral heterogeneity, price variations, and unique characteristics of Indian energy systems such as captive coal mining and non-market biomass consumption.

Multiple scenarios including Business-as-Usual (BAU), 2°C, Sustainable 2°C, and 1.5°C pathways are simulated for the years 2030 and 2050 to assess trade-offs between economic growth, employment, and emissions reduction. The findings highlight that while decarbonisation is feasible, it involves significant structural transformations, including increased reliance on renewable energy, improvements in energy efficiency, and policy-driven technological transitions. However, constraints such as market imperfections, structural inertia, and socio-economic disparities pose challenges to rapid transitions.

The study concludes that integrating climate policies with development strategies is essential for achieving sustainable growth in India. The hybrid modelling approach provides a more realistic representation of energy-economy interactions and offers critical insights for designing policy frameworks that align decarbonisation goals with socio-economic development.

Summary of the Thesis

This research investigates the complex relationship between economic development and climate change mitigation in the context of India, a rapidly growing economy with increasing energy demand and significant dependence on fossil fuels. The study is motivated by the global commitments under the Paris Agreement and the Sustainable Development Goals, which emphasize the need to achieve sustainable development while limiting global temperature rise. India faces a dual challenge of sustaining high economic growth and simultaneously reducing greenhouse gas emissions, particularly given its reliance on coal-based energy systems and the socio-economic imperative of poverty reduction and energy access. The research aims to contribute to this discourse by developing a comprehensive analytical framework that integrates economic growth, energy systems, and environmental constraints.

The primary objective of the study is to address methodological and analytical gaps in existing literature by developing a hybrid modelling framework that combines macroeconomic and technological perspectives. It seeks to evaluate the feasibility and trade-offs of achieving low-carbon growth in India under alternative development pathways while examining the alignment between decarbonisation targets and sustainable development objectives, particularly economic growth and employment generation. Additionally, the study explores the evolution of renewable energy policies at the global level to derive insights relevant for the Indian context. **A key contribution of the research lies in identifying critical gaps in existing approaches, including the lack of integration between energy system details and macroeconomic structures, the absence of microeconomic constraints such as market imperfections and structural inertia, inconsistencies between Input-Output tables and energy balance data, and insufficient representation of India-specific energy system characteristics**.

To address these gaps, the study adopts the IMACLIM-IND hybrid modelling framework, which integrates a top-down Computable General Equilibrium model with a bottom-up energy system model (AIM/Enduse). This approach allows for a detailed representation of both economic structures and technological processes, enabling a more comprehensive analysis of energy-economy interactions. A significant methodological innovation in the study is the construction of a hybrid Input-Output table, which reconciles physical energy data with monetary economic data. This process incorporates important features of the Indian energy system, including energy price heterogeneity, captive coal mining, non-market biomass consumption, and distortions in natural gas trading. By ensuring consistency between energy balances and national accounts, the hybrid table enhances the robustness and reliability of the modelling framework.

The study employs scenario-based analysis to explore different pathways for India's energy and economic future. Four scenarios are considered: Business-as-Usual, a conventional 2°C scenario, a Sustainable 2°C scenario with a phased reduction of coal use, and an ambitious 1.5°C scenario aligned with global climate targets. These scenarios are evaluated for the years 2030 and 2050, focusing on key indicators such as GDP growth, sectoral output, employment, and emissions. The findings indicate that decarbonisation is technically feasible but requires significant structural transformation of the economy. This includes a transition towards renewable energy sources,

improvements in energy efficiency across sectors, and substantial policy support to facilitate technological change. The results also highlight the existence of trade-offs, particularly in the short term, where stringent climate policies may affect economic growth and employment in carbon-intensive sectors. However, in the long term, these transitions can lead to more sustainable growth patterns and reduced environmental risks.

The analysis further underscores the critical role of policy instruments in driving the energy transition. Mechanisms such as feed-in tariffs, renewable purchase obligations, and carbon pricing are found to be effective in promoting renewable energy adoption and reducing emissions. At the same time, the transition process is constrained by several structural challenges, including market imperfections, infrastructural limitations, dependence on fossil fuels, and socio-economic impacts such as job displacement in traditional energy sectors. These constraints highlight the need for carefully designed policies that not only promote decarbonisation but also address issues of equity, employment, and social welfare.

Despite its contributions, the study acknowledges certain limitations. The modelling framework is largely static in nature, providing snapshots for specific time horizons rather than capturing continuous dynamic transitions. Data limitations, particularly with respect to renewable energy cost structures and non-market energy consumption, also affect the precision of the analysis. Additionally, the study relies on several simplifying assumptions related to capital accumulation, sectoral aggregation, and energy cost structures, which may not fully capture real-world complexities. Uncertainty in future technological developments and policy trajectories further adds to the limitations of scenario-based analysis.

In conclusion, the research demonstrates that decarbonisation and economic development can be pursued simultaneously, but this requires an integrated approach that aligns climate policies with broader development objectives. The hybrid modelling framework developed in this study provides a more realistic and policy-relevant tool for understanding the interactions between energy, economy, and environment in India. The findings emphasize the importance of strong institutional support, technological innovation, and international cooperation in achieving sustainable development goals. Future research can build on this work by incorporating dynamic modelling approaches, integrating climate impact assessments, conducting regional-level analysis, and including social dimensions such as inequality and employment transitions. The integration of advanced analytical techniques, including machine learning and stochastic modelling, also presents promising avenues for further exploration.

Chapter 1

Introduction

International unanimity to deal with climate change resulted in adoption of Paris agreement by more than 190 countries in December 2015. The agreement recognizes the Intended Nationally Determined Contributions (INDCs) that most countries submitted to the United Nations Framework Convention on Climate Change (UNFCCC), under which they have defined explicitly their respective targets for GHG emissions mitigation. It sets out the ambitious objective of "holding the increase in the global average temperature to well below 2°C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5°C above pre-industrial levels, recognizing that this would significantly reduce the risks and impacts of climate change".

Another major development in the year 2015 where more than 190 nations across the world made global commitments was the adoption of 17 sustainable development goals (SDGs) in UN Sustainable development summit in New York. In this research we focus on some of the SDGs adopted like SDG7 (ensuring access to sustainable energy for all), SDG13 (taking action to combat climate change) and SDG8 (promoting sustainable economic growth) focus on decarbonization of energy systems, GHG emission mitigation and high sustainable economic growth respectively. These SDGs underscore the linkages between decarbonization and development.

It is now acknowledged that low carbon pathway should be accompanied with meeting the development goals. Cancun agreement (2011) brought about a paradigm shift in climate change negotiations by emphasizing on creating a low-carbon society that ensures high growth and sustainable development. Moreover, the emission pathways are not only driven by climate change policies but also by the non-climate policies. So the articulation between climate and non-climate policies is imperative to develop appropriate action plans.

Developing countries like India face challenges quite different from that of the developed world since enormous development needs have to be balanced with the emission reduction targets. Large-scale poverty is one of the formidable challenges in the sustainable development pathway for India. India's INDCs specify the goal of increasing the share of clean energy in total energy consumption to 40%, which will contribute in achieving the target of reducing the emissions intensity by 33-35% by 2030 using the baseline of 2005. Meeting these climate change targets along with development targets is not straightforward as is also acknowledged in the INDC document submitted by India.

Our research focus is on improving the diagnostics about the policy questions on mitigation pathways in Indian context. Our work involves setting up the IMACLIM model for India and then answering key policy questions on aligning decarbonization of energy systems and sustainable development in India using the IMACLIM modelling tool.

1.1 Indian context: Energy – Environment – Economy

Developing countries have quite different context compared to the developed countries. Further significant differences exist even amongst the developing countries in terms of their energy, GHG emissions and economic situation. It is imperative to note the particular characteristics of energyenvironment-economy situation in India in order to delineate the climate issues that we will analyse through hybrid modelling.

India is one of the fastest growing economies of the world. It supports 18% of the world population on 2.4% of the world surface area. Energy is the key to meet the rapidly growing population demands on one hand and support economic development on the other. 75% of the demand for energy in India is met by fossil fuels with coal forming the backbone of Indian power sector accounting for 70% of power generation (India energy outlook report, 2015). Though India is the fourth largest emitter of Greenhouse Gases (GHG), the per capita emissions are much smaller compared to other major economies. Under the business-as-usual scenario, emissions are expected to rise in future.

Figure 1.1 shows how the indicators related to the objectives of decarbonization and sustainable development have changed overtime for Indian economy. The base index is taken to be 100 for the year 1990 for all variables for ease of comparison. Along with the rising population, GDP of India has been increasing. Also, there is a continuous growth in electricity generated, primary energy consumption and primary energy consumed per capita. Now CO2 intensity, which is carbon dioxide emitted per unit of energy consumed has not changed much since 1990's as the CO2 emitted has been growing at the same pace as energy consumed.

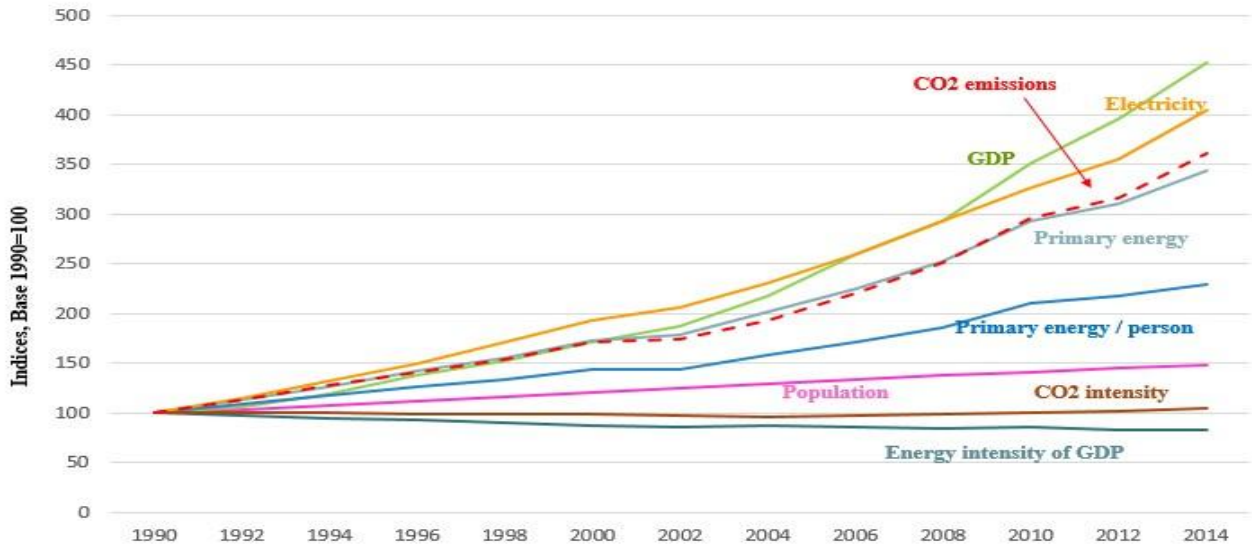

Figure 1.1 Development transitions in India

Source: World Bank, US EIA historical statistics, Indiastat, CMIE, The shift project data portal

India has signed the Paris agreement, formally committing to contributing in the efforts to limit the temperature rise above pre-industrial levels to below 2°C. India has committed to reduce the emission intensity of its GDP by 33-35% by 2030 from 2005 level and to scale up the renewable capacity by 5 times (Government of India (GoI), 2015). For achieving these targets, GoI has outlined a number of missions under the National Action Plan for Climate Change (NAPCC). Further, government has committed to SDGs and for this purpose has mapped all its policies to each SDG. The vision of the policy makers in India is influenced by the Father of Nation, Mahatma Gandhi when he says, "earth has enough resources to meet people's needs, but will never have enough to satisfy people's greed". We discuss the Indian energy-emission-economy profile and the policies made by the government in the subsequent sections.

1.1.1 Indian energy sector

India's growing population and urbanization will lead to an increase of energy demand by one-third by 2022 in INDC scenario compared to the demand in 2013 (IEA, 2015b) which will be met by fossil fuels to a large extent.

Indian energy sector is mainly coal based with most of the power being generated using fossil fuels. Over the recent years dependence on coal, oil and gas imports has increased (India energy outlook report (IEA), 2015). This can create issues for energy security in future, if adequate policy measures are not taken to increase the domestic production of energy sources. Further lack of power infrastructure can be gauged from the fact that 300 million Indians do not have access to residential electricity and many others get intermittent supply.

India's INDCs propose increasing the green power generation capacity to 175 GW by 2022, which includes the target of 100 GW solar power and 60 GW wind power. Number of other programs like National Smart Grid Mission, Smart Cities Mission, Make in India campaign[1] (zero defect zero effect- www.makeinindia.com) and green energy corridor projects have been rolled out with the aim of promoting sustainable development. There is a greater emphasis on energy conservation and on fuel efficiency standards. In India, subsidies on liquefied petroleum gas (LPG) and kerosene might go away in future as the deregulation of diesel and petrol prices in last 10 years. Further nuclear share of electricity generation is expected to go up. According to the results obtained from INDC scenario (IEA, 2015b) fossil fuels will account for 75% of the total energy demand in 2022. Electricity demand will double by 2030 in INDC scenario while renewable based power will triple compared to that in 2013.

1.1.2 Indian economy and GHG emissions profile for India

India is a developing economy with a growth rate of 7.6% in 2015-16 and per capita income of $ 7783 in 2018 at current prices (IMF, 2018).

Apart from dealing with the two major issues, rising energy demands and large-scale poverty, there is an urgent need to achieve low carbon growth. Since about 70% of the total GHG emissions comes from energy sector (IEA, 2015a), it is important to focus on energy sector for achieving the mitigation targets. India has the fourth largest energy-related CO_2 emissions in the world after China, the US and the European Union (figure 1.2).

CO_2 emissions are significantly high in India due to the dependence on fossil fuels for most of its energy needs. However per capita emissions are 10% of that in US and 25% of that in China. There are concerns that economic growth will lead to higher emissions.

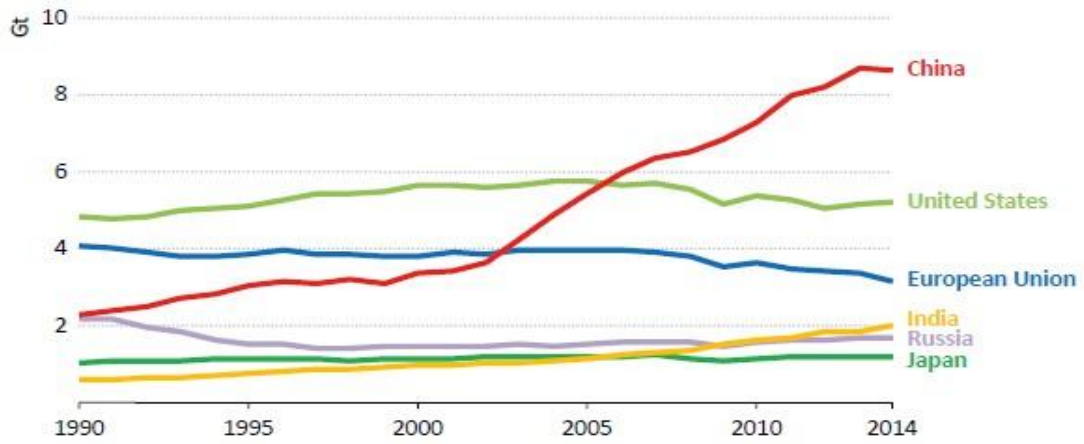

Figure 1.2 Energy related CO-2 emissions for different regions

Source: Energy and climate change report (IEA, 2015b)

1.2 Literature review

The links between decarbonization and sustainable development have been analyzed in multiple studies.

Table 1.1 Summary of literature review

Links between Decarbonizatio n & SDGs	**Energy systems**	Bottom-up model (TIMES; AIM/enduse)	-Decarbonization faster in sustainable development -Role of energy efficiency and renewables critical -Endemic energy system changes needed for low carbon growth	Vishwanathan et al. (2018); Byravan et al. (2017); Gambhir (2014) pathway growth
		Partial equilibrium model (GCAM; PROMETHEUS)	-Carbon price driving the technology penetration -Large emission gap between NDCs and 2°C pathways	Fragkos and Kouvaritakis (2018); Chaturvedi & Shukla (2014); Shukla & Chaturvedi (2012)
	Macro-economy	Top-down models (CGE; AIM/CGE; GEM-E3)	-Higher economic growth if mitigation actions embedded in development decision -Green policies can contribute to growth in short term, potential downsides for poor -India needs international support to achieve emissions reduction aligned with global temperature goal	Mittal et al. (2018); PC (2014); Weitzel et al. (2014); Saveyn (2012); Ojha (2009)
	Energy economy	Coupled models (AIM/Enduse & GEM-E3; Global climate model & CGE)	-GDP growth rate falls in carbon tax scenario -Energy efficiency improvement vital in buildings sector	Vishwanathan (2019); Pradhan & Ghosh (2012); Shukla et al. (2008)
Renewable Policies	**Policy framewor k**	Case studies, financial modelling	-Price and quantity policy instruments	IEA (2016), Renewables Global Status Report (2015), Shrimali et al. (2013)
	Evaluatio n of policies	Econometric model using panel data	-Feed-in Tariff (FIT) policy highly effective	Jenner et al. (2013), Dong (2012), Schmid et al. (2012), Couture et al. (2010), Butler et al. (2008)

1.2.1 **Decarbonization**

Decarbonization theme can further be studied under two focus areas: strategies for mitigation and renewables. In the first focus area that is strategies for mitigation, studies emphasize the importance of much more stringent action for mitigation than the currently proposed actions in order to bring Indian emissions to levels compatible with a global 2ºC target (Sluisveld et al.,2013). Other studies have suggested that shifting to renewable energy, negative technologies like carbon capture and storage and energy efficiency technologies in end-use sectors like households and industries are critical for achieving mitigation (WEO India report, 2015; Shukla & Chaturvedi, 2012).

1.2.1.1 ***Financial requirement***

Paris agreement has put a binding obligation on the developed countries to provide financial support to developing and underdeveloped countries and calls for voluntary contribution from other countries to the cause of emission mitigation. The goal of contributing US $100 billion per year until 2025 has been set for the developed countries. Government of India has set up a dedicated National Clean Environment Fund for mobilizing finances for clean energy by imposing coal cess. Total collection of US $ 2.7 billion is used for executing clean energy projects worth US $ 2.6 billion. The coal cess in this case can also be seen as carbon tax. Other fiscal measures taken by the government include deregulating petrol and diesel prices leading to price increases that would further contribute in emission reduction (India INDC document, 2015). 1.2.1.2 Technology transfer

Government of India had laid increasing emphasis on developing mitigation technologies in India through Make in India campaign. For this, international collaboration in disseminating knowledge on developing such technologies would be required in order to achieve quick results for GHG mitigation. Moreover since coal-based power capacity is expected to increase by 70% in INDC scenario (IEA, 2015b), efficient technologies in coal plants would be needed. Technology transfer from the developed countries or better utilization of domestic innovation abilities can be useful.

1.2.1.3 *Energy transition issues*

Electricity sector contributes about 40% of global Green House Gas (GHG) emissions (International Energy Agency, 2015). Coal power plants are known to be the largest contributor in the GHG emissions from power sector. Since 2000, the share of coal in energy related CO2 emissions has increased from 38% to 44% (IEA, 2015). The emissions from coal-fired plants can be reduced by either replacing the plants with gas power plants or with clean energy options like solar or wind power. Other options are to wait for their retirement or to decommission them earlier to switch to clean energy faster.

For meeting the electricity demands of millions of citizens and to fulfil infrastructure needs, India has seen huge expansion in its power capacity. Most of the capacity needs are being fulfilled by fossil fuels. India has enunciated in the INDCs that the priority will be increased deployment of

renewables and improvements in energy efficiency. However, even the rise in renewable share is not expected to spell the end of coal-fired power plants (figure 1.3).

renewables will become market competitive energy sources. Thus once the share of renewable power generation reaches a cut-off point, which can be determined through empirical analysis, the support policies like FiT will have to be withdrawn so that public exchequer is not burdened. Apart from this, there are integration challenges also for instance, smart grids have not yet been developed which can facilitate smooth transition.

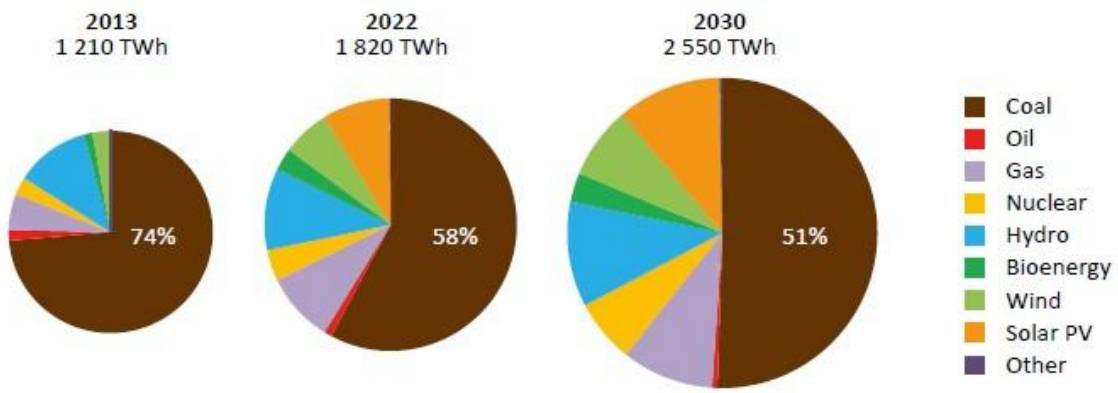

Figure 1.3 India electricity generation by type in the INDC scenario

Source: IEA report on Energy and climate change, 2015

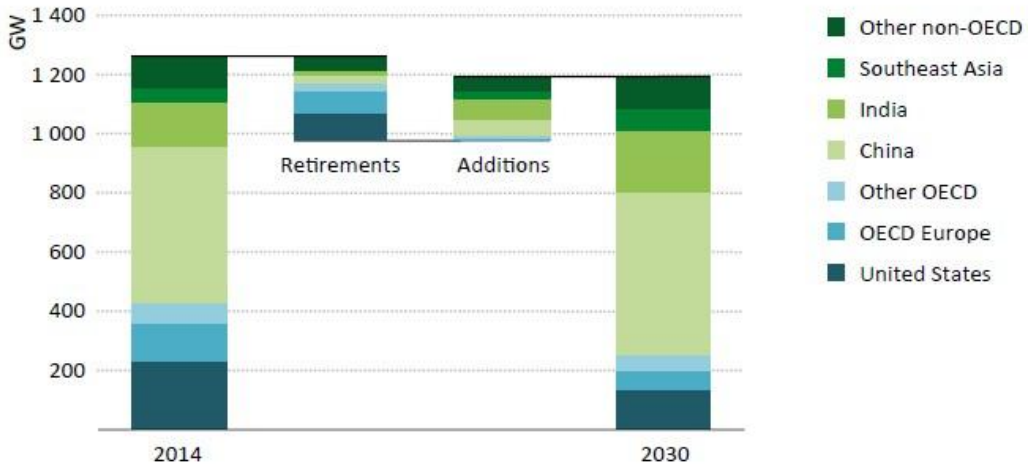

Figure 1.4 Subcritical coal capacity and retirements/additions in the INDC scenario by region

Source: IEA report on energy and climate change, 2015

There are social costs involved in energy transition like loss of jobs in polluting industry and this affects poor the most since they have neither access to markets, nor the skills to shift to another industry. So the policies need to provide support to the declining industries in the transition stage like allowing the time and providing finances to retrain and shift the employees to non-polluting jobs. Further this would also cause a clash of interests of two groups: Coal-based power generators and renewable energy producers.

1.2.2 Sustainable development

Sustainable development is at the core of energy and climate change policies in India. Figure 1.5 shows the timeline of selected development policies in India, which have direct interface and relevance with emissions, energy and climate policies. Since the policies driving the green revolution in 1970s to the most recent policies, there has been continued emphasis on crafting policies aimed at achieving sustainability goals. Also, it is evident from the mapping of policies to SDGs (Niti Aayog, 2015) that development is a critical aspect of climate and energy policies in India.

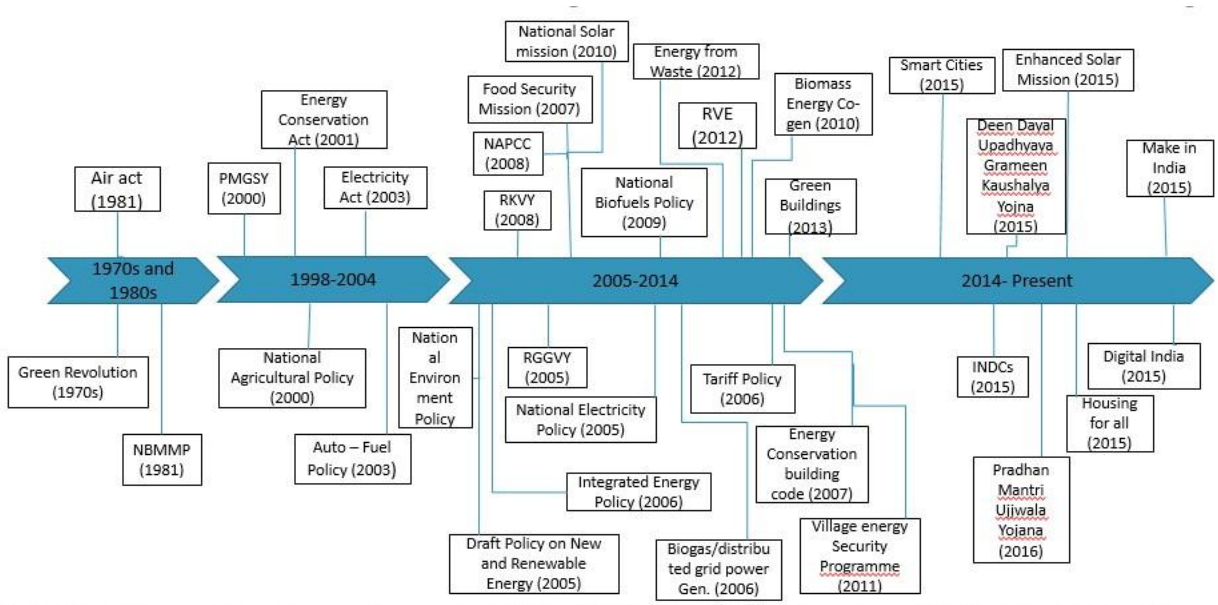

Figure 1.5 Timeline of development policies having interface with energy and climate policies

Source: MoP&NG (GoI), MoEF, MNRE, Ministry of Power, Ministry of Coal, Shukla et al., 2015

1.2.3 Aspects of energy systems

This theme deals with aspects specific to Indian energy systems like captive coal mining and nonmarketable biomass. The phenomenon of captive coal mining, which was introduced in the year 1993 by the government, implies that coal is being produced by sectors like power, iron and steel and cement for their own use. The idea behind allowing private companies to own coal mines was to boost thermal power generation in order to meet the increasing power demand. Though the percentage of captive coal (12%) is not significant compared to total coal produced, it is expected to grow in future (Coal Controller's Organisation, 2015).

1.2.4 Renewable policies

IEA has stressed the importance of renewable energy in mitigating the climate change (Olz, 2008). All 27 member states of European Union (EU) have implemented support policies for promoting renewable electricity (Jager et al., 2011). It should be noted that though renewable energy provides many advantages, it might pose some risks in terms of intermittent nature of Renewable Energy (RE) sources especially solar and wind and the requirement of some raw materials for RE generation, which are scarce.

Many policy instruments have been used in promotion of renewable electricity. Amongst the market based instruments there are two categories, price-based and quantity-based instruments. Fixed feed-intariffs (FIT) is a price-based instrument wherein price for renewable electricity is set exogenously by the regulator. Feed in tariffs are the guarantee of payment given by the state regulator to the renewable power generator for the electricity they produce. Energy utilities are supposed to buy electricity from RE producers at tariffs fixed by the regulator and additional burden on utilities is passed onto the power consumers by means of imposition of levy on end-use electricity prices or government supports the utilities by providing incentives. FIT can be technology-specific as well, wherein tariffs are determined based on cost of technology used for RE generation. Other instruments include investment grants by the government, subsidies. Amongst quantity-based instruments most significant is tradable renewable

energy certificate (REC) wherein electricity distribution utilities are supposed to buy some quota of electricity generation pre-specified by the regulator from the RE sources. In case the utilities buy renewable electricity in excess of their quota, they can sell their green certificates, at rates determined by the market. Another quantity-based instrument, tendering system allows for greater cost efficiency since once the tender is declared for providing certain amount of electricity using certain technology, bidding takes place which ensures that cheapest deal is received.

The effectiveness of these instruments has been determined in some studies based on econometric analysis using the experience in United States or EU (Butler et al., 2008; Couture et al, 2010; Jenner et al. 2013; Olz, 2008). In case of RECs the utilities chose the cheapest option to cover their quota, however in FIT tariffs are a function of the cost of technology used for generating RE, which allows diversification in technology. Technology penetration fostered by FIT is required in order to attain

the advantages of cost reduction due to learning effects. The problem of excess cost burden or free rider can be dealt by applying digression on the tariffs over the years. Both systems support innovation either due to competition (REC) or for higher profits (FIT).

FIT is more cost-efficient if it is technology specific rather than fixed since former would aid in limiting the producer surplus compared to surplus in latter case. REC offers other advantages like exposing RE sector to electricity market from the beginning and facilitating integration with other schemes like emission trading (Olz, 2008). In comparison, FIT has less market orientation but it offers investment security to private players. Both the schemes have been successfully implemented in different parts of the world with FIT dominating in EU and REC in US where it is more popular by the name Tradeable Green Certificates (TGC) (Jenner et al., 2013).

FIT policy instrument has been found to be highly effective amongst various price and quantity based policy instruments used across the world for promoting deployment of renewables (Dong, 2012). In 2015, feed-in policy was being implemented 73 countries (REN21, 2015) making it the most widely used policy instrument.

Government of India has been implementing several market-based policies for promoting renewables since 1990's. From the price-based instruments like feed in tariffs introduced in 2000 government has implemented quantity-based policies like renewable purchase obligation in 2011. The ambitious policy goals coupled with rapid progress in increasing the share of RE by India has made the country rank fifth in renewable energy attractiveness (E&Y report, 2015). Policy instruments as applied in India are marketbased instruments like FITs and fiscal incentives and control instruments like Renewable Purchase Obligations (RPOs). FIT for wind energy was introduced in early 2000 while for solar energy it was introduced in 2010 at central level through Jawaharlal Nehru Solar Mission (JNNSM) (Shrimali et al, 2013). States implemented preferential tariffs for solar energy in 2011. In Table 1.2 given below various policy instruments for solar and wind energy practiced in India are outlined.

The landmark regulation that transformed the power sector was Electricity Act implemented in 2003. It brought about changes like deregulating power generation, introducing open access in power transmission and permission to state electricity regulatory commissions (SERCs) to fix the RE obligatory procurement. In 2005, National Electricity Policy was formulated and National tariff policy in 2006 followed. These policies were made with the aim of increasing the RE share in total energy mix. Some of the policy instruments outlined in Table 1.2 are mutually exclusive like FITs and RECs or GBI and AD, others like financial incentives, wheeling charges are complementary to other policies like FIT and GBI.

Table 1.2 Policy instruments for renewable energy promotion in India

Policy instrument	Description
Feed-in Tariff (FIT)	Minimum prices at which RE is purchased from generating companies through long contracts (20-25 years) with transmission/distribution companies or trading licensees. FITs are determined by SERCs (except for JNNSM).
Generation based incentive (GBI)	Introduced in 2009 with incentive of Rs 0.50 per unit generated, however discontinued in 2012; GBI has been reintroduced in 2013
Renewable energy certificates (REC)	Market based instruments used to aid the companies meet the requirement under Renewable Purchase Obligation (RPO); introduced in 2011; entities can chose one of the two options between FIT or REC.
Fiscal incentives (excise, wheeling charges)	Concessional rates provided for excise (reduced from 8% to 0%) and customs duty for biomass, wind and solar. Wheeling charges for RE reduced as well.
Accelerated depreciation (AD)	GoI allows depreciation of 80% for renewable projects in first year; introduced in mid 1990s but discontinued for wind energy in 2012
Income tax exemption	100% tax waiver allowed on profits from RE power generation project; introduced in 2002 but expired in 2013.
Clean Development Mechanism (CDM)	RE companies can participate in Certified Emission Reduction (CER) credits market after getting registered with Ministry of Environment and Forests; introduced in 2005.

Source: Various government documents and news items

The changes in government support can be understood with the help of this diagram shown in figure 1.6. It represents a learning curve of renewable technology deployment. As the installed cost of renewables approaches that of fossil fuels price parity, the government support should reduce since the renewables now become competitive with fossil fuels in the market. One needs to be careful here as the fossil fuel prices are now quite dynamic.

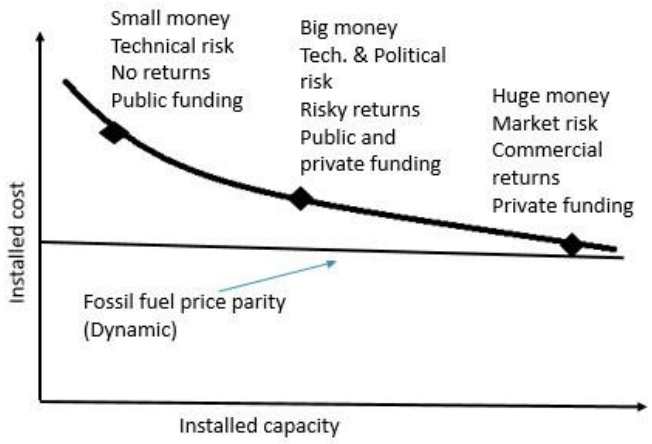

Figure 1.6 Stages of government support

Source: Adapted from Grubb et al., 2004

1.3 Research gaps

Input-Output (IO) tables are often used for studying the links between decarbonization and sustainable development. However, the issue is that the national accounts do not provide a complete description of energy systems. IO table describes the sale and purchase relationships between the producers and consumers within an economy. Energy balance on the other hand is the statistical accounting of production, transformation and consumption of energy products like coal, crude oil and natural gas. Since the energy balance provides a more reliable and disaggregated picture of energy systems, the IO tables of India need to be reconciled with energy balance using the energy prices in order to incorporate the energy systems description (Combet et al., 2014; Ghersi et al., 2006). Also, the issues specific to energy systems in India like captive coal mining, non-marketable biomass and natural gas trading need to be properly treated while reconstructing the IO table. The IO table thus reconstructed by modifying the table values is called a hybrid IO table.

Decarbonization and its links with sustainable development have been studied in the past using different approaches for India (Shukla et al., 2015; IPCC AR5 WGIII report, 2014). These studies do not incorporate the microeconomic constraints like market imperfections, structural inertia and myopic investment behavior (Ghersi, 2015; Hourcade, 2006)). Also, the studies do not capture the details of energy systems fully.

The research gaps based on the literature review discussed in the previous section can be summarized below in Table 1.3.

Table 1.3 Summary of research gaps

Themes	Literature	Research Gap
Methodology: Aspects of energy systems	Ghersi (2015), Combet et al. (2014), Weitzel et al. (2014), Ojha (2009), Mitra (1994)	• Energy price heterogeneity, dual accounting of volumes and values • Consistency of energyeconomy information in modelling framework
Renewable energy	Renewables Global Status Report (2015), Dong (2012), Schmid (2012), Hass et al. (2011)	• Global analysis of changes in renewable policies over time
Decarbonization of energy systems and sustainable development	Shukla et al. (2015), IPCC AR5 WGIII report (2014), Weitzel et al. (2014), Shukla et al. (2008)	• Capturing aspects of energy systems • Analysis with constraints such as market imperfections, structural inertia and myopic investment behaviour • Coupling top-down and bottomup approaches via iteration

1.4 Research questions

First question addresses the first research gap which is methodological in nature. We incorporate consistent information on India's energy-economy in the modelling framework. This is an important step in answering critical questions on India's energy-economy-environment systems. Third research gap pertaining to links between decarbonization targets and sustainable development is addressed through the second and third question given below. Fourth question addresses the second research gap of analyzing the changes the renewable policies across the world.

1. What are the methodological gaps in exploring the feasibility and trade-offs for achieving India's development and mitigation pathways? What is the way to overcome them?
2. What is the feasibility and trade-off involved in achieving high growth in a low carbon Indian economy in medium and long term based on the alternate growth dynamic based scenarios?

3. What would be the implications of aligning decarbonization targets and SDG 8 (decent work and economic growth) in future for India under alternate scenarios, such as, BAU, global decarbonization targets – 2ºC, 2ºC Sustainable, 1.5ºC?

4. How have policy instruments evolved for promoting deployment of wind based renewable electricity generating technologies across the world, and what policy lessons could be drawn for India?

1.5 Research framework

We propose to use the research framework shown in Figure 1.7 below for answering the research questions. We first capture the aspects specific to energy systems in India like captive coal mining, nonmarketable biomass and natural gas trading in the national accounts for policy evaluation. One important output of this process is hybrid input-output table (see chapter 2), which we will use to calibrate our model of Indian development & mitigation pathways. Renewable energy is one of the key component in the strategy for decarbonisation, hence we undertake a separate analysis of global renewable policies. Considering the bandwidth of work for this thesis, we narrow down our analysis to wind amongst all renewable technologies. This analysis provides useful inputs for defining the scenario drivers to be fed into the hybrid model developed for answering other two questions.

The model used is a hybrid model that is top-down model coupled with bottom up model (see chapter 4). Traditionally two kinds of models have been used for analyzing energy policies. Bottom-Up (BU) models emphasize on description of technological aspects of energy systems while Top-Down (TD) model focusses on macro-economic description of the whole economy and microeconomic constraints like imperfect markets or myopic investment behaviour. The two approaches need to be combined for analyzing links between decarbonization and SD as shown in Figure 1.8 below. Hybrid model which we intend to set up combines both these approaches. We use the hybrid model IMACLIM-IND/AIM India where BU AIM/Enduse and TD IMACLIM are coupled. The details of IMACLIM-IND model are discussed in chapter 4.

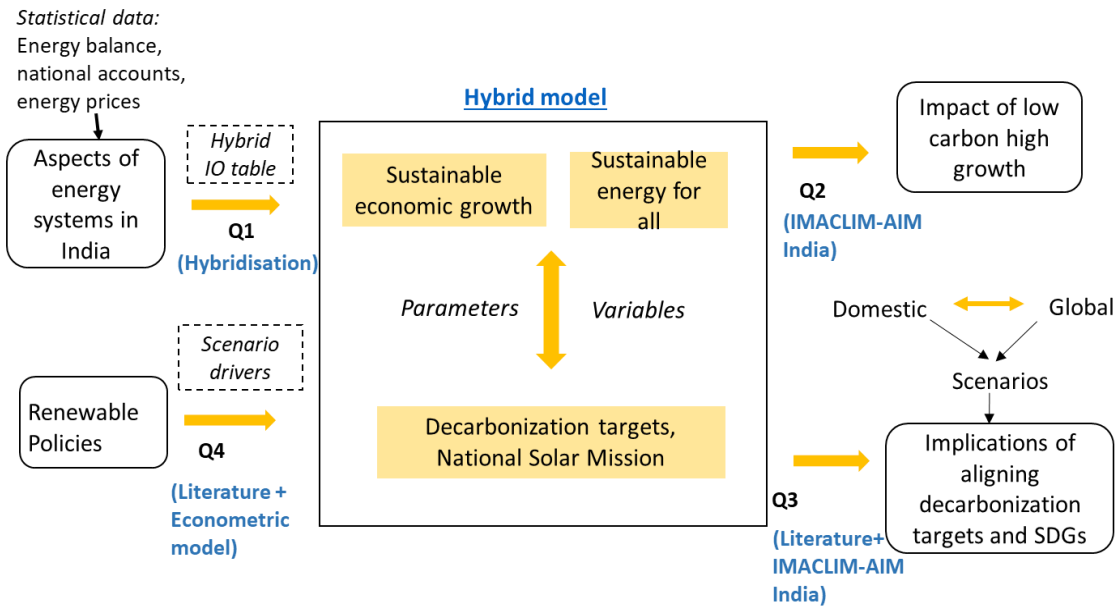

Figure 1.7 Research framework

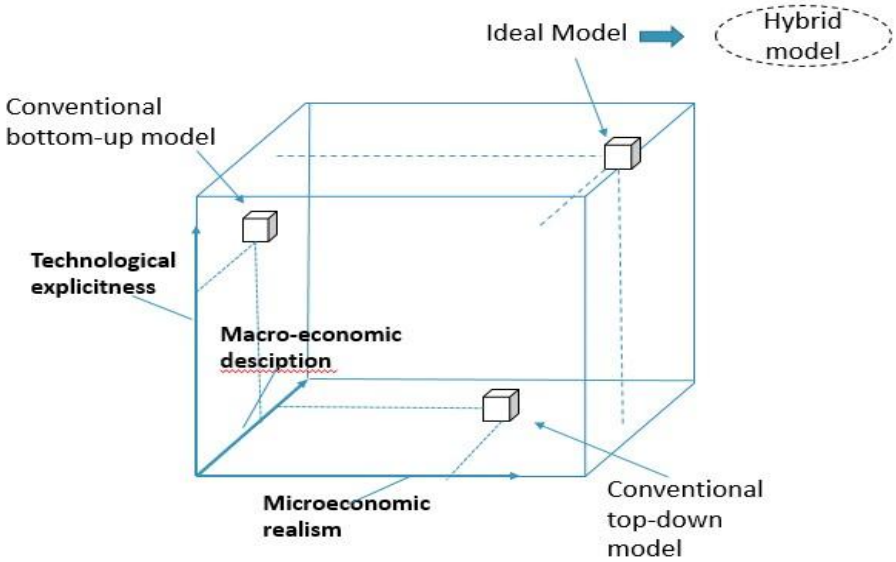

Figure 1.8 Categorization of energy-economy models

Source: Hourcade et al., 2006

1.6 Methodology and data sources

This section illustrates the research methodology we intend to use for this study. Our purpose is to explore development pathways under energy and carbon constraints which requires economy wide, hybrid multisectoral analysis. This is why we implement IMACLIM-IND model. We analyse the outlooks at specific target years, 2030 and 2050. Our model is not recursive with annual time steps rather it is static version giving the outlooks for specific horizons. However, we wish to control the macroeconomic dynamics of these medium and long term outlooks. To do so, we implement a 2-sector KLEM model which is a simplified sketch of 22-sector IMACLIM-IND. We first provide a brief overview of the model set up for this research (details can be seen in chapter 3). 2-sector KLEM model is based on standard
Computable General Equilibrium (CGE) model of 'KLEM' nature based on two primary factors of production capital (K) and labour (L) and two goods that is energy (E) and non-energy goods (M). The model and the results are discussed in the next chapter. The 2-sector model provides a macroeconomic picture and the flexibility to explore various test cases for understanding the macroeconomic dynamics clearly. KLEM allows us to explore structural change issues and secondary income distribution between aggregate economic agents in much simplified manner and this further helps in the setting up of 22-sector IMACLIM-IND model discussed in chapter 4 to articulate the climate policies in the context of Indian energy-economy system.

This economy-wide model is coupled with bottom-up AIM/Enduse model, which has already been setup in Indian Institute of Management (IIM), Ahmedabad. The two models are made to converge by iteratively exchanging the possible set of variables or parameters (Figure 1.9). IMACLIM runs under the constraint of exogenous data from AIM like energy import prices, energy output like coal, natural gas, crude oil and energy consumption of households. It computes various economic indicators like output of different sectors like cement, iron & steel, aluminium. These values are again sent back to bottom-up model. The output of different sectors drives the energy demand in bottom-up model for new simulation.

Compared to state-of-the-art modelling, the novelty resides in the specific features of the IMACLIM approach, namely: (i) a fully hybrid take on energy systems modelling combining bottom-up and topdown approaches, not limited to energy supply but extending to energy demand; built on a documented reconciliation of economic and energy statistics, for some recent year; and linking with BU modelling with due account of feedback loops *via* an iteration to convergence method. (ii) strong attention to

'second-best' constraints impairing energy transitions, as market imperfections, structural inertias, myopic investment behaviour or macroeconomic investment constraints.

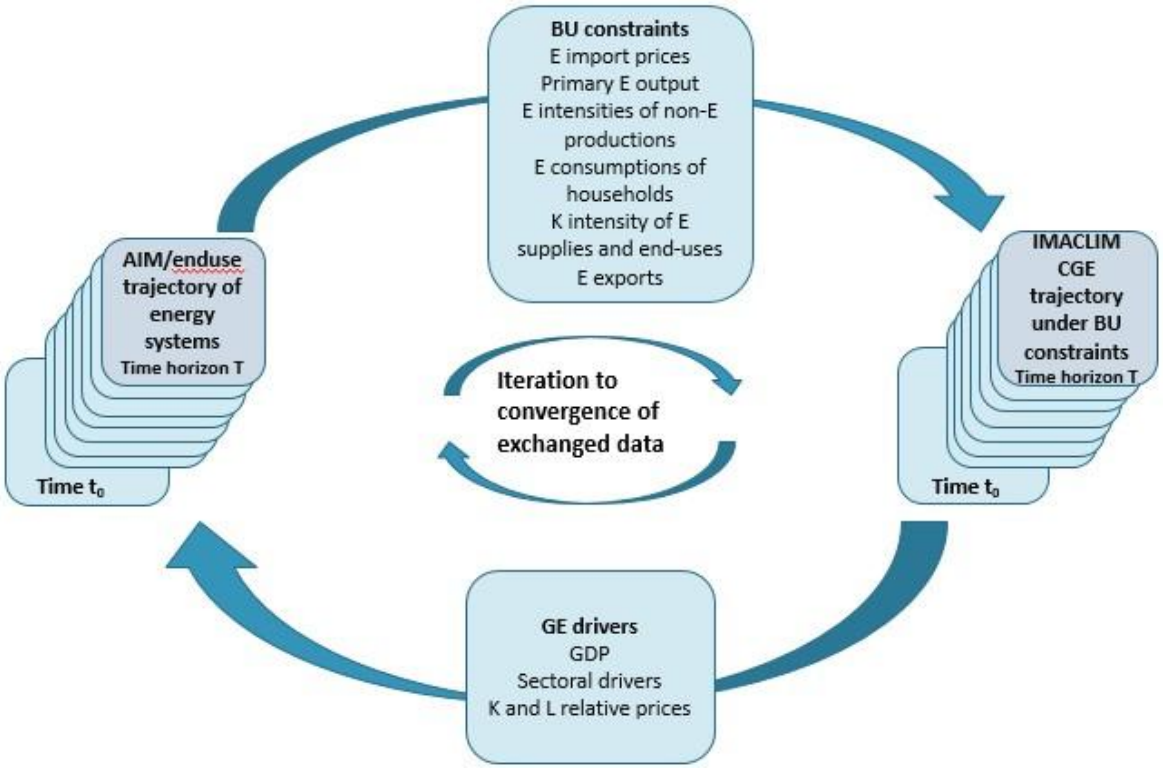

Figure 1.9 Coupling of TD and BU models for setting IMACLIM-AIM India hybrid model

1.6.1 Research question 1

The methodology to address Research question 1 is discussed in chapters 2, 3 and 4. It also involves soft coupling (Figure 1.9). Hybrid IO tables is developed for India by collecting the relevant data using distinct methodology. The tables thus developed provide three distinct advantages over the traditional methods. First, energy price statistics is used as opposed to previous modelling approaches, which derive energy prices from national accounts. Second, price heterogeneities faced by different economic sectors and households are reflected in the energy statistics. Third, in order to balance the accounts, necessary trade–offs are made by adjusting the variables.

Based on the recently released input output table for Indian economy, 2012 is chosen as the base year. The study involves 8 energy sectors and 14 non-energy sectors listed in Table 1.4 below. These sectors have been chosen based on the characteristics of energy systems in India.

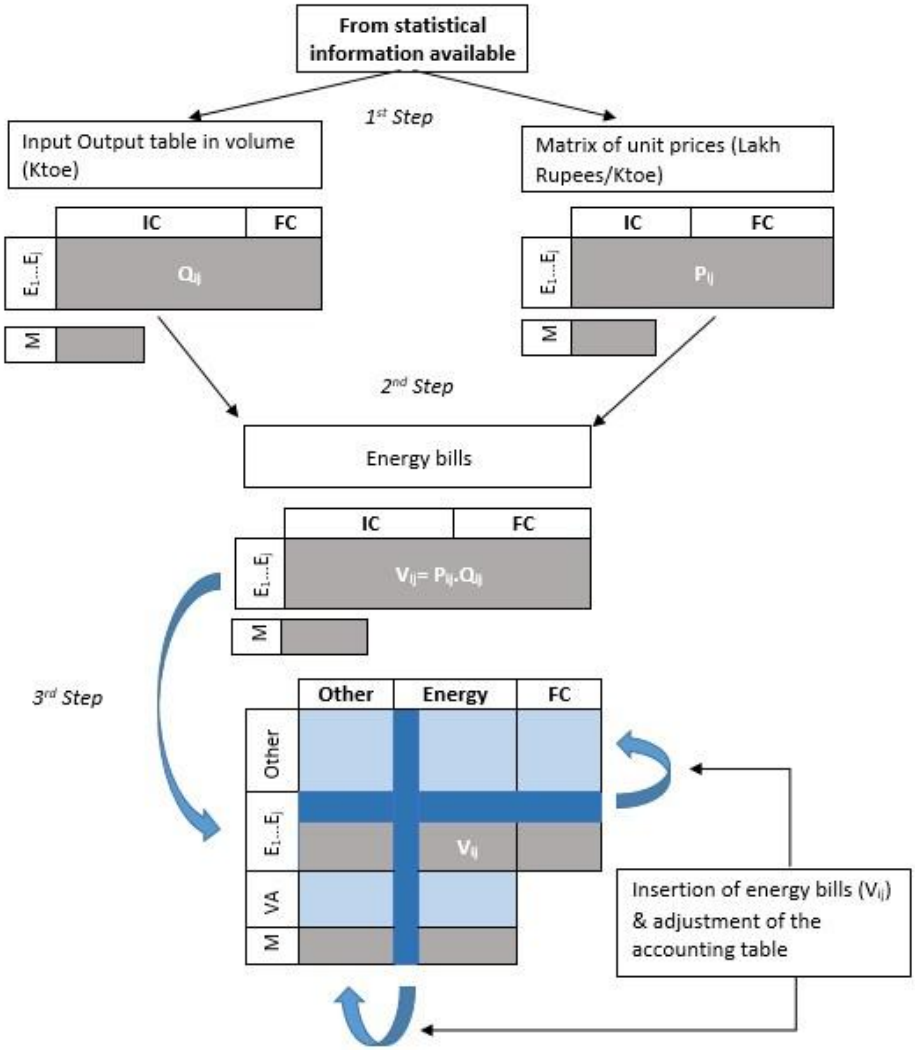

Figure 1.10 Schematics of the hybridization procedure

Table 1.4 Sectoral breakdown of final hybrid matrix (22 sectors)

Energy Sectors (8)		**Non- energy sectors (14)**	
COA	Coal	I&S	Iron & Steel
COK	Coke	CHE	Chemical
ONT	Crude oil and non-transport fuels	ALU	Aluminium
TRF	Transport fuels	CEM	Cement
BIO	Biomass	CON	Construction

GAS	Natural gas	TEX	Textile
ELE	Electricity	IND	Residual industries
REN	Renewable Energy	AGR	Agriculture
		ATR	Air transport
		WTR	Water transport
		ROA	Road transport
		RAI	Rail transport
		HOU	Housing
		SER	Other services

1.6.2 Research question 2 & 3

Methodology for question 2 and 3 is discussed in chapter 4. Hybrid IO table from Q1 and data on taxes, subsidies, employment and other variables are fed into the IMACLIM-IND model. On the other hand, energy system details are fed into AIM/Enduse. Now various scenarios are constructed for the model based on scenario drivers like the developments in socio-economic sphere, technology and environment. Literature review is done for building the scenario storylines. The method also involves sensitivity analysis of the key parameters used for scenario building. The two models are made to converge after iteratively exchanging the variables. Consistent with the model, the results are obtained for the years 2030 and 2050 for variables like GDP growth, sectoral growth, unemployment rate and production price growth.

(a) Business As Usual scenario (BAU). Indian INDCs of reducing emission intensity by 33-35% by 2030 from 2005 level and the slew of measures like promoting clean energy, enhancing energy efficiency, developing sustainable urban centers and sustainable transportation, controlling pollution, managing waste and enhancing carbon sink by increasing the forest cover.

(b) Conventional 2-degree scenario

(c) Sustainable 2-degree scenario – Here coal in power sector phases out by mid-century.

(d) 1.5-degree scenario - Recent Paris agreement at Conference of Parties (COP) 21 under the United Nations Framework Convention on Climate Change (UNFCCC) has taken a leap in terms of climate change mitigation regimes. It sets up an ambitious target of hitting 1.5°C rise in global temperature compared to pre-industrial levels. There is not much data available on the implications and emission pathways for 1.5°C target. Development of this scenario helps fill this gap.

1.6.3 are the year dummies (excluding 2009 period). SGEN is share of wind generation in total power capacity, WTP is wind turbine price and EP is electricity price. Z are other covariates like share of wind capacity, share of wind generation and absolute wind generation. U is the error term and α_i are the country specific unobservable effects.

1.6.4 Data sources

The study draws upon multiple data sources. Data is collected from secondary sources mostly like household surveys, national and international reports, and socio-economic statistics. Table 1.5 below provides the list of data sources.

Table 1.5 Sources for collection of data

Data	Sources
Energy balance	IEA, IIMA inventory
Carbon emission coefficients	IPCC reports, NATCOM
Energy prices	Ministry reports, companies' balance sheets, Indiastat
IO table	Central statistical Organization
Taxes and subsidies	Ministry reports, Indiastat
Employment data, household consumption	NSSO
Feed-in tariffs	IEA policy reports, Countries' energy policy documents
Renewable power	US EIA Historical statistics, World Bank, IEA

Chapter-2

Hybridization

The data hybridization process outlined below is the first step towards building an original EnergyEnvironment-Economy (EEE) modelling capacity for determining Indian mitigation pathways. Accounting matrices constructed in the past were mostly aggregated in nature with hardly any energy systems details. Full energy sector disaggregation in social accounting matrix was attempted by Deb Pal et al. (2012) for undertaking energy policy analysis. However, this method does not take into account the heterogeneity in energy prices, dual accounting of energy system in monetary value and physical units, and specific aspects of Indian energy systems like natural gas trading, captive coal mining and nonmarketable biomass consumption.

The hybridization process is based on two guiding principles for maintaining consistency of data. First, both physical and money values should follow the conservation principle that is resources and uses must be balanced. Second, the physical and money flows are linked by a unique system of prices implying that the money values can be obtained by multiplying the volumes by the corresponding price. Further, there are two rules guiding the methodology: one, the economic size is always preserved while correcting the statistical gaps; second, the purchasing price heterogeneities faced by different sectors and households is taken into account.

2.1 Methodology

The methodology followed to create the hybrid table has been documented in Combet et al., 2014. It unfolds in three main steps:

(1) Reorganizing the original energy balance data (in kilo tons of oil equivalent, Ktoe) and energy prices (in Lakh rupees/ktoe) into the sectoral distribution matching the input-output table (IOT) from national accounting. This not only involves reallocation of physical energy flows of energy balance to production sectors and households, but also entails re-interpretation of the flows in national accounting terms. In other words, it involves sorting out the flows that indeed correspond to economic transaction between national accounting agents. For instance, attributing the autoproduction of electricity to the accounting sectors; considering only the commercial flows especially in case of energy industry own use in energy balance; adjusting the data on international bunkers since energy balance reports data based on geography while IOT reports data based on national accounting rules.

(2) Multiplying the volumes with corresponding prices to obtain energy expenses at the same level of disaggregation as IOT.

(3) Plugging of the matrix of energy expenditures into original IOT and adjusting the other values of the table such that accounting approach is not disturbed and the total value added of domestic production remains same. This is done by: first, adjusting difference in uses and corresponding resources for energy sectors to the non-energy expenses on pro rata basis; second, by adjusting the difference in original and recomputed expenditures for the non-energy sectors to the most aggregated non-energy good which is 'other services' mostly.

The commodity X commodity Input Output (IO) table was constructed for 65 commodities using the recently released supply use tables for the year 2012-13 by the Central Statistical Office (CSO), the government organization responsible for coordinating statistical activities in the country. The IO table was constructed by manipulating the supply use matrix, with 140 products and 66 sectors (CSO, 2016), based on industry technology assumption. The data on energy volumes was taken from IEA and AIM end/use model set up at IIMA. The sources for heterogeneous prices for energy goods are specified in the excel sheet.

The decision regarding energy and non-energy sectors was taken based on the specific features of Indian energy sectors and Indian economy. For instance, we take cement and aluminium manufacturing sectors since these are the two most energy intensive sectors in the Indian economy. Government of India (GOI) has specified these sectors as the focus areas for meeting the energy efficiency targets for instance in the policy named Perform Achieve Trade (PAT) scheme. Further decision to add renewable sector was based on the policy framework being pursued by GOI wherein the target has been set to achieve 175 GW renewable energy capacity by the year 2022 (Union Environment Ministry India, 2015).

Figure 2.1 Final sectoral breakdown

Energy Sectors (8)	Non- energy sectors (14)
Coal	Iron & Steel
Coke	Chemical
Crude oil and non-transport fuels	Aluminium
Transport fuels	Cement
Biomass	Construction
Natural gas	Textile
Electricity	Residual industries
Renewable Energy	Agriculture
	Air transport
	Water transport
	Road transport
	Rail transport
	Housing
	Other services

2.2 Indian energy systems aspects

Some notable aspects of Indian energy systems discovered in the hybridization process and their treatment are described below.

- The coal expenses going into electricity sector in original IO were just 25% of those obtained by multiplying available price and volume estimates (hereafter the 'volume x price' approach). The official documentation on IO reveals that the coal expenses have been calculated using the inputs of electricity distribution companies like state electricity boards, departmental commercial undertakings of central and state governments and private electricity companies. On the other hand, the volumes in energy balance have been computed using the coal controller's reports which give the numbers for the output of coal companies going into electricity generation sector. Literature shows that thermal efficiency of coal plants in India is 30% on average (Colin, 2015).

- Another source of difference in IO and volume x price coal expenses is the phenomenon of captive coal mining (introduced in the year 1993) implying that coal is being produced by sectors like power, iron and steel and cement for their own use. The purpose of the government in allowing private companies into coal mining is to boost the thermal power generation in order to meet the increasing power demand.

- In order to treat the goods properly, the costs of captive coal mining must be transferred to the coal sector, which is actually the sum of coal mining activities regardless of which sector undertakes the activity. The process involves following steps: (1) the coal expense of captive mines operators is increased via a price x volume approach using the appropriate coal cost net of profit as the price; (2) all cost elements of the coal mining 'sector' (activity) are increased homothetically in order to rebalance rise in sales; (3) the costs of the captive mine operator for each item are reduced such as to exactly compensate the cost increase in the coal mining activity. The broad idea is to transfer the costs of the captive coal mining to the general coal mining activity, and to treat captive coal expenses as any other coal expenses. The question of an increase of the share of captive mining in coal expenses can be taken care while modelling pathways by assuming a decrease of the average profit rate of coal mining.

- Next is the trading issue that is natural gas being bought by the refined petroleum sector to be sold to consumers. The refined petroleum products expenses going into chemical and electricity sector from original IO is 2 and 1.5 times respectively of the expenses obtained by volume x price approach. On the other hand natural gas expenses (original IO) into electricity sector is just 0.3% of the expenses from other approach. Natural gas expenses into chemical sector (original IO) are 37% of those obtained from volume x price approach. Refined petroleum products sector appears to play a role of trader, buying a huge amount of natural gas and selling it back to other businesses without consuming it. This implies that the switch from an industry x industry to a commodity x commodity matrix is not complete, there remains some natural gas sales covered by the refined pet products 'sector' of the commodity x commodity matrix. In such a case IO values can be misleading, hence we decide to use volume x price data.

- Bulk of the total energy consumption by households in India is for cooking purpose. Biomass such as firewood, cowdung and agricultural residues, which are normally collected by the households themselves (Pachauri, 2007) is the most commonly used fuel for this purpose. The data on household

expenditure on biomass is obtained from the National Sample Survey Office (NSSO), which conducts regular socio-economic survey. Though we have an estimate of the monthly per capita expense on firewood and cow-dung and percentage of people using these fuels, it is hard to get an estimate for the non-market consumption that is number of people collecting the biomass themselves. We compare the household expenses on forestry products specified in IO table (1.3% of total household expenditure) with the firewood and cowdung (1.83% and 0.16% of total household expenditure) consumption from NSSO data. The two numbers seem compatible considering the fact that some proportion of biomass is non-marketable. Hence, we decide to take IOT household expense on forestry sector for representing household expenditure on biomass in the final hybrid matrix.

- Another noteworthy issue was the fact that there are significant amount of non-energy uses of some petroleum products like petroleum coke, lubricants, naphtha and other non-specified oil products in India as opposed to the situation in developed economies. While bitumen non-energy uses were adjusted in the construction sector, petroleum coke was adjusted in cement sector since it is one of the largest consumers of petroleum coke in India. Remaining petroleum coke and other non-energy uses of petroleum products were distributed on pro-rata basis of the respected unaccounted share in volume x price compared to IO expense across all sectors.

- Considering the increasing prominence of renewables in the Indian energy policies and the specific tariffs and incentives for this sector, it was added as a separate sector in the matrix. The costs into renewables sector were assumed proportional to the electricity costs after deducting the fossil fuel costs. This assumption is taken for lack of more data on the cost structure. This will be refined later by looking into the share of capital costs in the cost structure of each form of renewables.

2-sector KLEM model is based on standard Computable General Equilibrium (CGE) model of 'KLEM' nature based on two primary factors of production capital (K) and labour (L) and two goods that is energy (E) and non-energy goods (M). CGE model specifies the economic relationship between different markets in the economy in mathematical terms. CGE model is useful in predicting the change in economic variables because of climate and energy policies. This model is based on Solow Swann growth model where yearly economic growth will be derived using exogenous assumptions of demography, productivity of primary inputs like capital and labour, and savings rate. At the bottom of the production structure shown in Figure 3.1, K & L are traded off through a constant elasticity substitution (CES) function to produce a KL aggregate. In second step KL and E are traded off to produce and in final step KLE and non-energy good is traded off to produce output Y.

3.3 KLEM model vis-a-vis IMACLIM-IND

The purpose of setting up 2-sector KLEM model is to set up a simple model to frame the macroeconomics of the multisector model. As opposed to the 2-sector model, in the multisector model we chose not to explore the explicit trajectory since we undertake analysis for single year horizon. Rule of explicit capital accumulation is used in 2-sector model based on annual depreciation rate assumption while we assume implicit capital accumulation in IMACLIM-IND as 2-sector is recursive dynamic model which provides results for 38 years from 2012-2050 while we run a static single year horizon model in case of multisector analysis. It should be noted that we can resort to the trick of implicit capital accumulation in

IMACLIMIND because we have already framed the macroeconomics of explicit capital accumulation with KLEM. The nested production structure is used based on CES functions in 2-sector model while in multisector we switch to Leontief production functions keeping the CES only for K and L substitution based on the reliance on engineering notion that is taking information on capital and energy intensities from bottom-up model.

The institutional division of money flow amongst four agents that is households, firms, government and rest of the world is not taken into consideration for 2-sector analysis. The equations used in the model are discussed below.

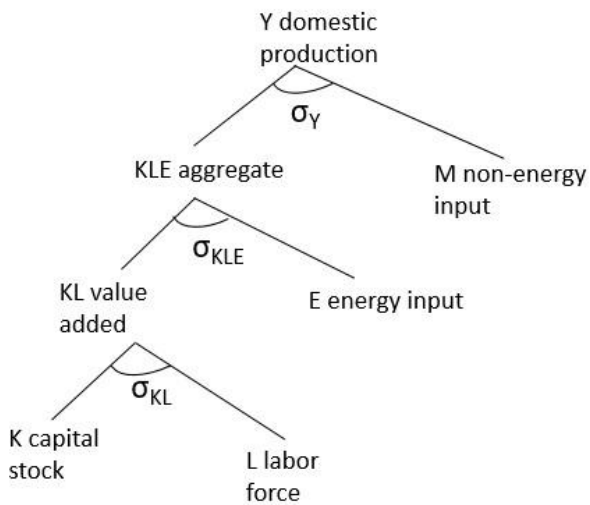

Figure 3.1 Nested production structure of non-energy good in 2-sector KLEM model

3.4 KLEM formulary

Before introducing the KLEM formulary, it will be useful to look at the KLEM model notations which will be used in the equations below. The table below provides the details of all the variables and parameters used in KLEM-India model. Some parameters remain constant based on calibrated values at the base year 2012 while others are taken from other sources. The model counts 43 equations listed in chapter 3 and 43 variables listed below. The equations (1), (14), (15), (17), (26), (28), (31) each count twice and equation (27) counts 4 times for intermediate goods price information thus making 33 equations count to 43. We do not use time index 't' except when necessary. Five-year time periods computed in the model except a 3-year period from 2012 to 2015 implying years analysed are 2012, 2015, 2020, 2025 up till 2050. We substitute index 'E' for energy goods aggregate and 'Q' for non-energy goods aggregate.

Notation	Description	Variables	Parameters
L	Labour endowment	0	1
K	Capital endowment	0	1
φ_L	Labour productivity (index 1 in 2012)	0	1
L_i	Labour going in good i production	2	0
K_i	Capital stock in good i production	2	0
KL	Value-added aggregate of K_Q and L_Q in the production of good Q	1	0
KLE	Aggregate of value-added KL and energy E in the production of good Q	1	0
E_i	Consumption of energy in the production of good i	1	1
C_i	Consumption of good i by households	1	1
G_Q	Consumption of good Q by government	1	0
I_Q	Investment in good Q	1	0
X_i	Export of good i	1	1
M_i	Import of good i	1	1
Y_i	Output of good i	1	1
S_i	Total supply of good i	2	0
α_{ij}	Intensity of good j in good i	2	2
λ_i	Labour intensity of good i	1	1
κ_i	Capital intensity of good i	1	1
w	Average wage	1	0
p_K	Rental price of capital	1	0
p_{KL}	Price of value-added KL in good Q production	1	0
p_{KLE}	Price of aggregate KLE in good Q production	1	0
p_{Y_i}	Output price of good i	2	0
p_{Si}	Average price of good i supply	2	0
p_{ij}	Price of good i used in the production of good j	2	2

p_{C_i}	Price of good i for households	1	1
p_{G_Q}	Price of good Q for government	1	0
p_{I_Q}	Price of good Q for investment	1	0
p_{X_i}	Export price of good i	1	1
p_{M_i}	Import price of good i	1	1
CPI	Consumer Price Index (chained Fischer index)	1	0
MPI	Import Price Index (chained Fischer index)	1	0
u	Unemployment rate	1	0
GDP	Gross Domestic Product	1	0
B	Trade balance	1	0
a_{KL}	Coefficient of K_Q in the KL CES function (calibrated in 2012)	0	1
b_{KL}	Coefficient of L_Q in the KL CES function (calibrated in 2012)	0	1
σ_{KL}	Elasticity of substitution between K_Q and L_Q in KL	0	1
a_{KLE}	Coefficient of KL in the KLE CES function	0	1
b_{KLE}	Coefficient of $\alpha_{EQ}\, Y_Q$ in the KLE CES function	0	1
σ_{KLE}	Elasticity of substitution between KL and $\alpha_{EQ}\, Y_Q$	0	1
a_Y	Coefficient of KLE in the Y_Q CES function	0	1
b_Y	Coefficient of $\alpha_{QQ}\, Y_Q$ in the Y_Q CES function	0	1
σ_Y	Elasticity of substitution between KLE and $\alpha_{QQ}\, Y_Q$ in Y_Q	0	1
σ_{wu}	Elasticity of real wage to the unemployment rate	0	1
σ_{Mp}	Elasticity to relative prices of the share of imports in the total non-energy resource	0	1
σ_{Xp}	Elasticity to relative prices of the non-energy export trend	0	1
δ	Depreciation rate of the capital stock (4%)	0	1
s_G	Ratio of public expenditure to GDP	0	1
s_I	Ratio of investment to GDP	0	1
τ_{ST_i}	Sales tax on good i sales	0	2
τ_{Yi}	Output tax on good i production	0	2
$\tau_{MS_{ij}}$	Specific margin on good i sales to good j production	2	2

τ_{MSC_i}	Specific margin on good i sales to households	1	1
τ_{MSX_i}	Specific margin on good i exports	1	1

3.4.1 Calibration

We compute the parameter values that are compatible with our base-year statistical information that is 2012 hybrid input-output table and all model equations.

Capital stock must be carefully calibrated to keep control on the unemployment trajectory produced by the model[1].

We define the base year stock K_0 as:

$$K_0 = I_{Q_0} \cdot \frac{A}{\delta_K + g_{av}}$$

which basically derives from the recognition that I_0 is a fraction of k_0 with:

- A is the ratio of the investment rate S_I (averaged over the period 2017 to 2050) to its 2012 value S_{I0}. I_0 is multiplied with A to capture the possible misalignment of the 2012 investment value with the average level over the projection period. We chose to focus on 2017 to 2050 investment rates to control the unemployment in the later years considering that it is constrained by dynamic calibration in the period before 2017.
- δ_K is the depreciation rate which is in the denominator to recognize that I_0 is supposed to entail replacement of the capital stock scraped at the end of period 0.
- g_{av} is the average annual growth rate (product of exogenous productivity and labour endowment) from 2012 to 2050, which is also in the denominator to acknowledge that I_0 is supposed to provide the growth of the capital stock for supporting the growth in economic activities with the simplistic assumption of constant capital intensity of production.

This method of estimating K_0 is used in 2-sector KLEM model while in 22-sector IMCLIM-IND model we consider another approach.

[1] Overestimating K_0 would dwarf calibration year investment I_0, which would not allow the capital stock to increase in pace with efficient labour endowment which in turn impacts on activity and hence employment. On the other hand, an underestimated K_0 will result in capital stock growing faster than the efficient labour, thereby reducing the relative cost of capital at the benefit of activity and employment.

3.4.2 **Production**

The production structure of energy goods does not involve nested trade-offs rather it is based on exogenous assumptions on energy supply description from AIM/Enduse model. The energy consumption in production of energy and non-energy goods is computed as follows:

$$E_i = \alpha_{Ei} Y_i \qquad (1)$$

According to the nested non-energy production tree, K and L are traded off at the bottom of the structure with constant elasticity of substitution σ_{KL} so the CES production function is: : $KL = (a_{KL}\, K_Q{}^{\rho_{KL}} + b_{KL}\, (\phi_L\, L_Q)^{\rho_{KL}})^{\frac{1}{\rho_{KL}}}$ where $\rho_i = \frac{\sigma_i - 1}{\sigma_i}$. L_Q is the labour mobilized augmented by the productivity factor ϕ_L. Using the prices p_K and p_L, cost minimization produces the following equations for non-energy good.

$$L_Q = \frac{1}{\phi_L}\left(\frac{b_{KL}\,\phi_L}{p_L}\right)^{\sigma_{KL}} \left(a_{KL}{}^{\sigma_{KL}}\, p_K{}^{1-\sigma_{KL}} + b_{KL}{}^{\sigma_{KL}} \left(\frac{p_L}{\phi_L}\right)^{1-\sigma_{KL}}\right)^{\frac{-1}{\rho_{KL}}} KL \qquad (2)$$

$$K_Q = \left(\frac{a_{KL}}{p_K}\right)^{\sigma_{KL}} \left(a_{KL}{}^{\sigma_{KL}}\, p_K{}^{1-\sigma_{KL}} + b_{KL}{}^{\sigma_{KL}} \left(\frac{p_L}{\phi_L}\right)^{1-\sigma_{KL}}\right)^{\frac{-1}{\rho_{KL}}} KL \qquad (3)$$

In case of composite production, α_{EQ} is exogenous hence cost minimization process is truncated and KL is deduced from KLE aggregate function which is a CES of KL and energy ($KLE = \left(a_{KLE} KL^{\rho_{KLE}} + b_{KLE} E_Q^{\rho_{KLE}}\right)^{\frac{-1}{\rho_{KL}}}$):

$$KL = \left(\frac{KLE^{\rho_{KLE}}}{a_{KLE_Q}} - \frac{b_{KLE_Q}}{a_{KLE_Q}} . \alpha_{EQ}{}^{\rho_{KLE}}\, Y_Q{}^{\rho_{KLE}}\right)^{\frac{1}{\rho_{KLE}}} \qquad (4)$$

On the next higher tier, *KLE* aggregate for non-energy good and composite inputs $\alpha_{QQ}\, Y_Q$, facing prices p_{KLE} and p_{QQ}, are traded off with a constant elasticity of substitution σ_{YQ} to form domestic production Y_Q. Cost minimization process induces the following:

$$KLE_Q = \left(\left(\frac{a_{Y_Q}}{p_{KLE}}\right)^{\sigma_{Y_Q}} . a_{Y_Q}{}^{\sigma_{Y_Q}} . p_{KLE}^{1-\sigma_{Y_Q}} + b_{Y_Q}{}^{\sigma_{Y_Q}} . p_{QQ}^{1-\sigma_{Y_Q}}\right)^{\frac{-1}{\rho_{Y_Q}}} . Y_Q \qquad (5)$$

$$\alpha_{QQ} . Y_Q = \left(\frac{b_{Y_Q}}{p_{QQ}}\right)^{\sigma_{Y_Q}} . \left(a_{Y_Q}{}^{\sigma_{Y_Q}} . p_{KLE}^{1-\sigma_{Y_Q}} + b_{Y_Q}{}^{\sigma_{Y_Q}} . p_{QQ}^{1-\sigma_{Y_Q}}\right)^{\frac{1}{\rho_{Y_Q}}} . Y_Q \qquad (6)$$

3.4.3 Final consumption and investment

While the household consumption of energy goods is exogenous taken from AIM/Enduse model, the nonenergy goods' household consumption adjusts based on endogenous/flexible value of household and foreign savings obtained as a result of fixed exogenous investment value taken from World Bank Development indicators (2018). This closure rule follows the Johansen rule (Sen, 1963) as opposed to the standard CGE closure rule of fixed domestic and foreign savings and investment computed endogenously.

Investment in (immobilization of) non-energy goods is exogenous share s_I of GDP. Investment in (immobilization of) energy goods is assumed to be zero (some stock variations in original IO were cancelled out in the hybrid IO).

$$p_{IQ}\, I_Q = s_I\, GDP \tag{7}$$

Government spending in non-energy goods is again a constant share of GDP while in energy goods is zero as per the national accounting conventions.

$$p_{GQ} G_Q = s_G GDP \tag{8}$$

GDP is defined as (computed from expenditure side):

$$GDP = \sum_{i=E,Q} p_{Ci} C_i + p_{Gi} G_i + p_{Ii}\, I_i + p_{Xi}\, X_i - p_{Mi}\, M_i \tag{9}$$

3.4.4 International trade

International trade of energy goods (X_E, M_E, p_{XE}, P_{ME}) is exogenous from IEA World Energy Outlook reports. With regard to the non-energy goods, the competition in international trade markets depends on the relative prices of goods. The contribution of trade to resources and uses i.e. the ratios of imports M_i and exports X_i to total supply S_i are elastic to the evolution of domestic (p_{Yi} or p_{Xi}) versus foreign (p_{Mi}) prices.

$$\frac{M_Q}{S_Q} = \frac{M_{Q_0}}{S_{Q_0}} \left(\frac{p_{M_0}\, p_{Y_Q}}{p_{Y_{Q_0}}\, p_{M_Q}} \right)^{\sigma_{Mp}} \tag{10}$$

$$\frac{X_Q}{S_Q} = \frac{X_{Q_0}}{S_{Q_0}} \left(\frac{p_{X_0}\, p_{M_Q}}{p_{M_0}\, p_{X_Q}} \right)^{\sigma_{Xp}} \tag{11}$$

The trade balance B is:

$$B = \sum_{i=E,Q} p_{Xi} X_i - p_{Mi} M_i \tag{12}$$

In the final baseline discussed in section 3.4 we chose specific closure rule. The ratio between the domestic and international vectors of prices does not result from a trade balance assumption as in standard CGE models. Rather, it is dictated by the assumption of an unemployment rate stabilised at base year level:

$$u = u_0 \tag{13}$$

3.4.5 Market clearings

The balance of goods markets is between supply, which comprise domestic production Y_i and imports M_i, and uses, which consists of households' and public consumptions C_i and G_i, immobilisations I_i and exports X_i. For energy goods, the data hybridization process results in this equation being expressed in kilo tons-of-oil-equivalent (Ktoe), consistent with the 2012 Indian energy balance of the International Energy Agency (IEA). The public consumptions and immobilisations of all energy goods are nil, by national accounting convention for the former and by definition for the latter.

$$S_i = \sum_{j=E,Q} \alpha_{ij} Y_j + C_i + G_i + I_i + X_i \,,\, i \in \{E, Q\} \tag{14}$$

$$S_i = Y_i + M_i \,,\, i \in \{E, Q\} \tag{15}$$

Total labour demand for both energy and non-energy sectors is:

$$L_i = \lambda_i Y_i \,,\, i \in \{E, Q\} \tag{16}$$

Where λ is the labour intensity. Labour market clearing requires that labour demand and unemployment balance out labour supply from households. The labour supply of households or labour endowment L is an exogenous parameter computed from International Labour Organisation (ILO) 2018 estimates until 2021 and considering the 2020 ratio of labour to working age (20 to 69) population (UN, 2015) from 2021 on.

$$\sum_{i=E,Q} L_i = (1 - u)\, L \tag{17}$$

A wage curve describes the correlation between the unemployment rate and the average wage

(Blanchflower and Oswald, 2005). The purchasing power of wages—the average wage at current prices w deflated by the consumer price index CPI—is elastic to unemployment variations $\frac{u}{u_0}$ with σ_{wu} elasticity, around a trend defined by labour productivity gains ϕ_L:

$$\frac{w}{CPI} = \phi_L \, w_0 \left(\frac{u}{u_0}\right)^{\sigma_{wu}} \tag{18}$$

Total capital write-offs from the non-energy sector are:

$$K_Q = \kappa_Q \, Y_Q \tag{19}$$

Where κ is the capital intensity.

For the energy sector, the constraint is applied to the productive capital stock that it is not allowed to contract faster than the assumed depreciation rate of 4% (n is the number of years between the previous and the current period).

$$K_{E,t} = max((1-\delta)_n \, K_{E,t-n}; \kappa_E \, Y_E). \tag{20}$$

The demand and supply of capital balance out:

$$\sum_{i=E,Q} K_i = K \tag{21}$$

Capital endowment K is a static parameter but a dynamic variable, which follows the standard accumulation rule starting from K_0 at calibration year, depreciating at rate δ and accumulating each year's physical investment flow I_Q.

3.4.6 Producer and consumer prices

Primary factor payments in terms of wages and price of capital are assumed to be same for both the sectors and they are computed based on the market balance equations.

Price of KL aggregate (p_{KL}) is a canonical function of labour and capital prices as KL is a CES aggregate of K and L.

$$p_{KL} = \left(a_{KL}^{\sigma_{KL}} {p_K}^{1-\sigma_{KL}} + b_{KL}^{\sigma_{KL}} \left(\frac{p_L}{\phi_L}\right)^{1-\sigma_{KL}}\right)^{\frac{1}{1-\sigma_{KL}}} \tag{22}$$

Since energy consumption values are taken from bottom-up model, the cost minimization process is truncated for computing p_{KLE} (price of KLE aggregate, which is specific to non-energy goods' production). We rather consider a simple accounting equation.

$p_{KLE} \, KLE = p_{KL} \, KL + p_{EQ} \, \alpha_{EQ} Y_E$ (23) The producer price p_{YQ} for non-energy good is again a canonical function derived after cost minimization process of CES production function using the price of KLE aggregate and non-energy input to total domestic production $\alpha_{QQ} Y_Q$. We need to add the constant calibrated output tax value τ_{YQ} and the mark-up rate τ_{MQ} (for profit margin) for getting the actual producer price.

$$p_{Y_Q} = \frac{\left(a_Y^{\sigma_Y}\, p_{KLE}^{1-\sigma_Y} + b_Y^{\sigma_Y}\, p_{QQ}^{1-\sigma_Y}\right)^{\frac{\rho_Y - 1}{\rho_Y}}}{\left(1 - \tau_{Y_Q} - \tau_{M_Q}\right)} \tag{24}$$

On the other hand, producer price for energy good is computed by adding the cost of all inputs:

$$p_{YE} = p_{QE}\, \alpha_{QE} + p_{EE}\, \alpha_{EE} + p_L\, \lambda_E + p_K\, \kappa_E + \tau_{ME}\, p_{YE} + \tau_{YE}\, p_{YE} \tag{25}$$

The imported non-energy goods' price is the *numéraire* of the model and the imported energy goods' price follows an exogenous trajectory computed using the information from IEA World Energy Outlook (2016).

We consider the imported and domestic energy goods to be homogenous rather than assuming Armington specification as commonly done in CGE models. This is to maintain an explicit account of physical volumes, which is one primary component of our approach of modelling.[1] For the sake of simplicity, we treat non-energy goods similarly to energy goods. The average supply price of good i, p_{Si}, is therefore the weighted average of the output and import prices of good i, with the output and import volumes as weights.

$$p_{Si} = \frac{p_{Yi}\, Y_i + p_{Mi}\, M_i}{Y_i + M_i} \tag{26}$$

The price of intermediate goods p_{ij}, i.e. the purchaser's price of good i for producing good j, is equal to the supply price augmented by specific margins that embody deviations of purchaser prices from the
average output price τ_{SMij}, ad valorem sales tax τ_{ST} and excise taxes t_{ETij}:

$$p_{ij} = (p_{Si}\,(1 + \tau_{SMij}) + t_{ETij})(1 + \tau_{VATi}) \tag{27}$$

The process of data hybridisation takes care of only the energy goods' margins, they are assumed nil for non-energy goods. For all goods and all years, we keep them at their calibration values.

The purchaser's prices of good i for households p_{Ci}, government p_{GQ}, investment p_{IQ} and foreign agents (export price) p_{Xi} follow definitions similar to that of intermediate prices, with the difference of the additional value-added tax at rate τ_{VATi} for all sales but exports, which do not either bear any excise tax.

$$p_{Ci} = (p_{Si}\,(1 + \tau_{SMCi}) + t_{ETCi})\,(1 + \tau_{VATi}) \tag{28}$$

$$p_{GQ} = (p_{SQ}\,(1 + \tau_{SMGi}) + t_{ETGi})\,(1 + \tau_{VATi}) \tag{29}$$

[1] The Armington specification differentiates the domestic and imported versions of goods. It creates a hybrid of both goods, whose volume does not proceed from the summing of those of the imported and domestic goods.

$$p_Q = (p_{SQ}\,(1 + \tau_{SMIi}) + t_{ETIi})\,(1 + \tau_{VATi}) \tag{30}$$

$$p_{Xi} = p_{Si}\,(1 + \tau_{SMXi}) \tag{31}$$

A chained Fisher consumer price index (CPI) measures the aggregate evolution of consumer prices. Because it is chained, it is a dynamic variable building each year on the price and consumption changes from the previous year:

$$CPI_t = CPI_{t-1}\sqrt{\frac{\sum_i p_{C_{i,t}}\,C_{i,t-1}}{\sum_i p_{C_{i,t-1}}\,C_{i,t-1}}\,\frac{\sum_i p_{C_{i,t}}\,C_{i,t}}{\sum_i p_{C_{i,t-1}}\,C_{i,t}}} \tag{32}$$

Similarly, the import price index is:

$$MPI_t = MPI_{t-1}\sqrt{\frac{\sum_i p_{Mi,t}\,M_{i,t-1}}{\sum_i p_{Mi,t-1}\,M_{i,t-1}}\,\frac{\sum_i p_{Mi,t}\,M_{i,t}}{\sum_i p_{Mi,t-1}\,M_{i,t}}} \tag{33}$$

The purpose of implementing the KLEM maquette is to produce insights on the macroeconomic consequences of the Indian energy constraint, which can be used to implement the full hybrid IMACLIM-AIM India model. The method to produce such insights consists of starting from a stable growth path (SGP) outlook, where all exogenous physical flows (energy consumptions, export trends, etc) follow the potential (sometimes also called 'natural') growth trajectory defined by the combination of labour supply and productivity gains. Then sequentially introducing AIM trajectories concerning energy flows and prices or costs, to elicit the impact of each successive assumption on the SGP. Next we illustrate various tests of systematically introducing the bottomup exogenous information into the model in order to investigate the macroeconomic behaviour from the model results.

3.5 Macroeconomic investigation

India exhibits a high trade deficit at its calibration year, of which as much as 87% is attributable to Energy. This explains why the dynamics of E trade weigh so much on the macroeconomics of any trajectory if we set the trade balance at some exogenous level. Whatever this level, international trade of the composite good must adjust to the dynamics of E trade (exogenous, drawn from AIM model), which happens via adjustment of the vector of domestic prices with respect to that of international prices—see below.

Index 1 = 2012

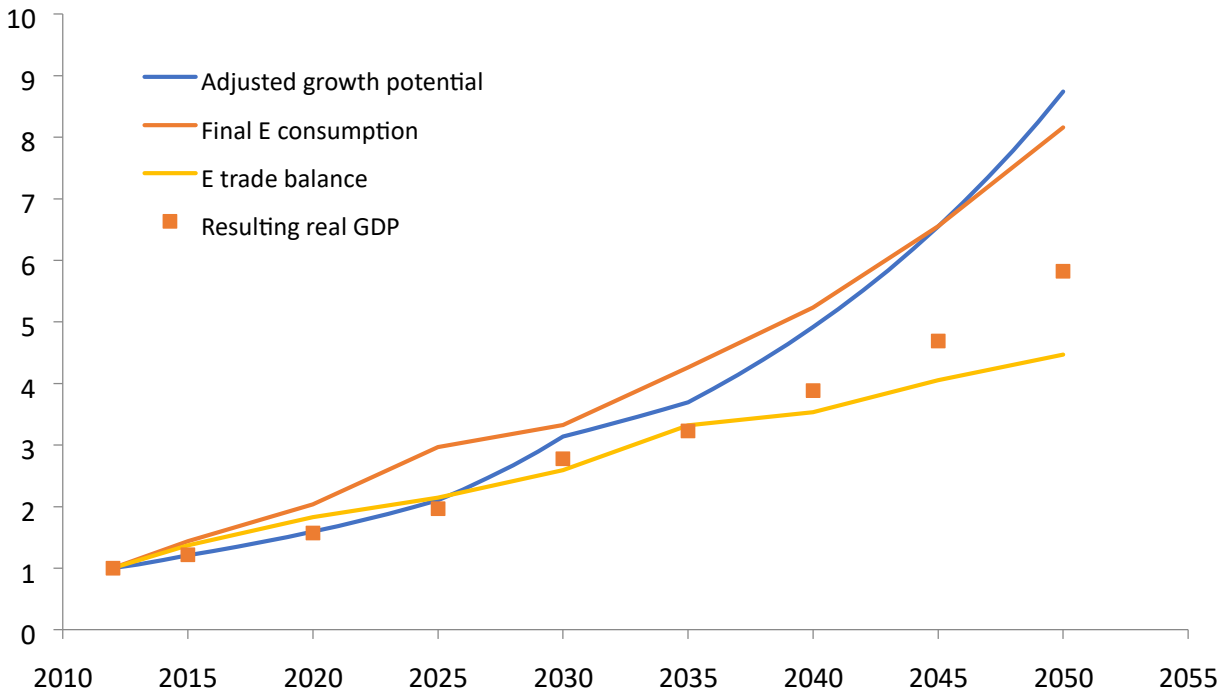

Figure 3.2 Growth potential vs. Final E expenses and E trade, India (from Test 15 in the descriptive below)

Sources: For growth potential, crossing of exogenous labour force and labour productivity assumptions; for E data, AIM/Enduse BAU scenario defined in Appendix B.

The baseline trajectories of E trade and E expenses drawn by AIM turn out to grow faster than adjusted potential value-added up to 2025 for the former and 2035 for the latter (Figure 2.2),[1] then slower.[2] This suggests that effective growth will be below its potential in the short run, then will possibly catch up.

1. Labour productivity gains for all energy goods' output

Here we assume that exogenous labour supply and productivity assumptions (6.17% AAGR - potential growth) extend to all E volumes and composite exports. We chose not to surmise productivity gains at the high rates forced on non-energy productions. We can ultimately derive insights from AIM on the evolution of O&M costs, but we lacked time to push model coupling so far and we leave it to further research. We find that the effective growth matches potential growth when labour productivity gains extend to E output (labour intensity of E at calibration value divided by labour productivity gain), as expected. There are

[1] 'Resulting real GDP' is just the plot of one specific run and can be ignored.

[2] By adjusted potential value-added we mean the potential level of GDP defined by the exogenous assumptions of labour productivity and labour force trajectories, taking account of the fact that labour productivity gains do not apply to the energy sector.

some minor fluctuations in prices and unemployment (see Figure 3.3) caused by the exogenous drivers implying that if the 2 drivers had constant yearly rates of increase then the match would be perfect.

2. No labour productivity gains for energy goods' output

Now we consider the case where productivity gains do not extend to E. Then the economy only achieves a 5.72% AAGR, close to half a point below the 6.17% AAGR of potential growth. This compounds into a 15% real GDP gap in 2050. The lower growth only barely induces higher unemployment. This is because energy trade volumes evolve as potential GDP and energy trade deficit tends to increase, whereas composite trade deficit tends to be stable because composite trade volumes evolve as real GDP; as a consequence, domestic prices are raised in order to reduce both the energy trade and composite trade deficits slightly so as to maintain trade balance stability. The rental price of capital decreases (compared to Test 1- Figure 3.3) because of the higher relative capital supply, which is induced by the use of potential growth in place of effective growth for the computation of initial capital stock. This implies that the rise in prices happens via labour prices. However, household energy consumption still evolves as potential GDP in volume. Energy production does not benefit productivity gains so the price of this consumption increases substantially because of skyrocketing labour costs driven by productivity increases. The total effect on purchasing power is a slight decrease, which further induces a slight increase of unemployment.

3. Adjustment of labour productivity assumption applied to energy volumes and capital K_0 estimation

We increase the energy volumes at 5.72% a year and the expected growth rate used in the computation of K_0 is at 5.72% as well. The resulting AAGR is 5.71%. Unemployment rises a bit more, reaching 4.15% in 2050, because domestic price rises are not anymore required to cancel the increased E trade deficit.

4. Exogenous energy trade volumes and trade prices

The results confirm that with the even lower energy trade balance of AIM (both in volume and prices) domestic prices reduce further in order to increase the composite trade deficit and thus reach the trade balance target. This induces higher unemployment, at 4.89% in 2050. The impact on growth is however quite limited and the AAGR remains at 5.71%.

5. Exogenous household energy consumptions and prices

From the previous test, we conclude that the domestic energy consumption assumptions (prices and volumes) have a strong influence on results related to unemployment and growth. We introduce AIM household energy consumptions in two steps first just the volume, and second both volume and specific margins (Test 5a) (Figure 3.3). The volume alone raises 2050 unemployment to close to 6%, while the AAGR drops to 5.66%. With AIM assumption lying far below the 5.72% AAGR assumption retained in Test 4, the reason for the unemployment hike is that household consumption switches from energy to composite consumption; but composite output is blocked by the trade balance assumption (because of the constancy of the ratio of imports and exports to output), which remains unchanged; and contrary to the

energy output to which it substitutes, composite output benefits productivity gains. As a consequence, the switch results in lower employment. This is quite counterintuitive result since lower household energy consumption induces lower growth and higher unemployment. This is the consequence of working in a framework where the energy and composite trade balance is exogenous at least in share of GDP (grossly speaking). Next, we add AIM's household E price, which is higher (Test 5b). The only consequence is that of limiting the budget transfer from energy to composite consumption, while not increasing in any way the amount of labour mobilised by E output. As a consequence, it induces a higher unemployment rate, albeit only marginally (increasing from 5.95% to 6.00%).

Potential growth: 6.16% (exogenous labor productivity and labor supply assumptions).
Unemployment set at 2012 calibration value

	Assumptions					
	Test1	**Test2**	**Test3**	**Test4**	**Test5a**	**Test5b**
E volumes	6.16%	6.16%	5.72%	AIM (X:3,M:3,P_X:1.1,P_M:1)	C_E:1.8	SM_{CE}
E output (Phi)	Yes	No	No	No	No	No
Growth for computing K_o	6.16%	6.16%	5.72%	5.72%	5.72%	5.72%

	Results for 2050					
	Test1	**Test2**	**Test3**	**Test4**	**Test5a**	**Test5b**
Effective growth	6.16%	5.72%	5.71%	5.71%	5.66%	5.66%
U (2050)	3.57%	3.77%	4.15%	4.89%	5.95%	6%
P_K (2050-2012)	1.6%↓	30%↓	18%↓	32%↓	32%↓	33%↓
P_{YQ} (2050-2012)	0%	3.6%↑	0.2%↓	11%↓	10%↓	10%↓
P_{CE} (2050-2012)	0%	36%↑	33%↑	20%↑		

Figure 3.3 Step-wise introduction of bottom-up information and the consequent results in KLEM model

6. Exogenous E prices and E consumption by non-E firms

We introduce further AIM's non-E firms' energy consumption trajectory, in volume and price simultaneously (the specific margin variations are too small to warrant specific scrutiny). The shock is massive. Unemployment explodes to 25%, while the AAGR is brought down to 4.34%. The mechanics at play are twisted. With E trade and E household consumption fixed, the collapse of the energy intensity of non-E production forces E output downwards. However, the exogenous E consumption of E firms remains set at the same level. Consequently, E prices increase massively, which causes the consumer price index to rise, whereas there is a downward pressure on wages because of the drop in E output. The real wage collapses under the combined two forces, which further induces a very high unemployment rate.

7. Exogenous E consumption by E firms

We now add AIM's trajectory for the E consumption of E production, for volumes first. Interestingly the 2050 unemployment rate only drops to 22.75%, which means that the story of Test 6 is not all in the irrelevant energy price hike—which is now contained because E consumptions of E production are consistent with the rest of E volumes. So the story really is similar to that for households. Lower energy consumptions do not benefit growth because they induce higher composite consumptions. But composite output is blocked by the trade balance constraint. Composite trade deficit is only 1%, and remains so (price variations neglected) because both imports and exports are proportional to output. If composite output were to increase when E output decreases, then the trade deficit will decrease (with the E trade deficit at 6.4% at calibration year). Note that the E trade deficit is set by AIM in absolute value as AIM provides both the 2 volumes and the 2 prices. This means that when GDP increases (measured in international composite good) the ratio of E trade deficit to GDP decreases, and vice versa, with impact on the composite trade deficit required to meet the trade balance constraint.

8. E consumption (prices) by E firms

This trajectory decreases by almost 33%, which means that the negative margins (akin to rebates) levied on energy sales to energy firms tend to decrease, i.e. prices of 'E in E' tend to increase. As expected, this price increase induces a decrease of real GDP and an increase of unemployment, although very slightly, because the initial specific margin at the calibration year is only of -2%.

9. GFCF trajectory (decreasing trend)

Now, we include the decreasing GFCF trajectory inferred from statistical observation over past years. This has little bearing on the results, but it unexpectedly increases real GDP and decreases unemployment, although very slightly. The main impact of the decreasing GFCF trajectory is to simultaneously decrease the initial capital stock and the investment trajectory. In fact, taking account of the GFCF trajectory to calibrate the initial capital stock cancels out any first order impact of modification in the GFCF trajectory. Remaining second order impacts are the consequence of the compounding effect of the investment change. The linear approximation of using an average GFCF rate in the initial capital stock computation underestimates the compounding effect of investment superseding its average rate for the starting years.

10. Exogenous non-E inputs (material, capital, labor)

Next, we add the only missing input from AIM, i.e. the trajectory of non-E inputs to E, which is meant to reflect changes in the average E output price not explained by price shifts of E in E. After a 27% drop in 2015 the trajectory of that cost item is steadily increasing up to a significant 2.552 factor, which means that non-E intensities of E end up exogenously increased by 150%. The impact is again massive, as unemployment climbs to 29.4% and GDP AAGR is reduced to 4.04%. The mechanism at play echoes that highlighted by Tests 5, 6 and 7 below: the unchanged E output mobilises 150% more input to deliver the

same service. In addition, labour input must shift from composite to energy production, where it does not benefit labour productivity gains.

We reach some preliminary baseline trajectory that integrates all relevant macro drivers and AIM trajectories, albeit under the constraint of a stable trade deficit trajectory and nil price elasticities of composite trade. It is therefore proportional to composite output in volume, and requires adjustments of the real effective exchange rate (the ratio of domestic to international prices) to accommodate the trade deficit assumption (Figure 3.4). This baseline trajectory is marked by a high unemployment rate and an AAGR 1.8 points below its value if taking into account the exclusion of energy output from labour productivity gains. It is also characterised by a marked stability of the composite output price, which remains within 5% of its initial value relative to the international composite price.

	Assumptions				
	Test6	**Test7**	**Test8**	**Test9**	**Test10**
E volumes/prices	Non-E firms energy consumption:3.2/prices	E firms E consumption:3.65	E in E prices: 1.34		Non-E intensities of E
GFCF	No	No	No	Yes	Yes
Price elasticities	Nil				

	Results				
	Test6	**Test7**	**Test8**	**Test9**	**Test10**
Effective growth	4.34%	4.56%	4.55%	4.56%	4.04%
U (2050)	25%	22.75%	22.88%	22.79%	29.4%
P_{YE}(2050-2012)	136%↑	68%↑	69%↑		
P_{YQ}(2050)(over test 5)	9%↑	7%↑	7%↑		
P_{CQ}(2050)(over test 5)	8%↑	6%↑	6%↑		

Figure 3.4 Step-wise addition of other AIM information

11. Price elasticities for trade (50%)

We introduce a 50% price elasticity of the composite import and export shares. Because prices were quite stable in previous case, the elasticities make little difference, but a negative one. They bring unemployment to 30.04% and the AAGR to 4.02%. The reason for the negative impact is that (1) having positive elasticities means that the slight 2050 domestic price decrease of Test 10 induces a decrease of the import share and an increase of the export share; (2) to compensate this improvement of the trade balance domestic prices must be contained—this has a negative impact on the balance because with price elasticities below 1 volume variations do not compensate price variations; (3) this has a negative impact on the purchasing power of wages; (4) the wage curve warrants that this decrease of purchasing power translates into an increase of the unemployment rate.

12. Trade elasticities (greater than one)

On switching the trade elasticities to 150%, the impact is still quite limited, but compared to previous case it is now positive (29.04% unemployment and the AAGR back at 4.04%, with in fact real 2050 GDP slightly above that of Test 10). The reason is that with elasticities above 1 any price adjustment induces a contrary volume adjustment of greater magnitude. As a consequence, to compensate the trend of improvement of the trade balance (compared to Test 10) domestic prices must now be increased, with converse impacts (compared to Test 11) on the unemployment rate and the AAGR.

13. Trade elasticities (100%)

In Test 13 we bring trade elasticities to 1, to confirm our interpretations above but that does not happen.

Unemployment reaches 28.98% and real 2050 GDP is above that of either Tests 10, 11 or 12 (Figure 2.5). This means that we are missing some additional mechanism, probably linked to the fact that our trade balance constraint is expressed as a ratio to GDP—whereas our reasoning above implicitly considers it in absolute terms. This would require more analysis to be fully explained. However, the weak sensitivity of model results to these elasticities in the context of a constant trade balance allows us to stop our investigation here.

	Assumptions		
	Test11	**Test12**	**Test13**
Price elasticity	50%	150%	100%
	Results		
	Test11	**Test12**	**Test13**
Effective growth	4.02%	4.04%	4.044%
U (2050)	30.04%	29.04%	29.98%

Figure 3.5 Results with varying trade elasticities

14. Trade balance (suitable value)

We now turn to the final stage of our exploration: the forcing of trade balance variations. Preliminary tests confirm the notorious high sensitivity of modelling results to such shifts. To begin exploring model sensitivity we simply adjust the end-year (2050) trade balance target of Test 13. With the 100%

elasticities of Test 13, decreasing the trade deficit ratio by 10% requires decreasing domestic prices, which implies decreasing the purchasing power of wages and hence increasing unemployment, by a substantial 1.3 points (to 31.2%). This is bad news, because it means that under our specifications so far (including

AIM baseline results) a scenario of decreasing trade deficit induces unsustainably high unemployment levels. The direction of the impact of constraining a reduced trade deficit is the same when explored in the context of Test 12, with 150% trade elasticities; however, it is opposite in the case of Test 11 with trade elasticities at 50%—which, satisfactorily enough, appear more acceptable than those above
1.

In Test 14 we therefore consider 50% trade elasticities and a trade deficit uniformly decreasing by 1% per year. The unemployment trajectory comes out close to its asymptote until 2025, then increasing up to 16% in 2050. This trajectory echoes the comparison between the exogenous potential growth and E trade balance trajectories. As a consequence, in another unrecorded test we stabilise the trade deficit up to 2025, then force it to decrease by 2% per year up to 2050. The end-year unemployment rate comes out at 13%.

15. Constant unemployment constraint

With this unrecorded test we in fact started a trial-and-error approach focusing on the unemployment rate trajectory, which might take long to complete. In Test 15 we change approaches and substantially alter the model by substituting a constant u constraint to the Trade balance constraint. Consecutively now endogenous trade deficit trajectory comes out quite acceptable, maintained around 7 to 8% of GDP up to 2030 then decreasing to 2.5% of GDP in 2050. Here, we end up with a domestic-to-international composite price ratio 4 times what it is in year 2012, which induces a REER 3.14 times above 2012. This can be translated in a massive Rupee appreciation. It is compatible with the REER projections in CEPII working paper (Fouré et al., 2012) which points at doubling of REER by the year 2040. Especially in the Balassa-Samuelson framework, Total Factor Productivity and energy productivity catch-up leads to real exchange rate appreciation at a pace proportional to the share of non-tradable goods in the country. The appreciation of the currency in developing countries derives from an increase in the relative price of nontradable commodities to the tradable ones and higher share of non-tradable commodities in the consumption basket. India is expected to see a strong real exchange rate appreciation up to 2050 (Fouré et al., 2012).

	Assumptions	
	Test14	**Test15**
Trade elasticities	50%	50%
Trade deficit	1% decrease per year	No
U	No	3.6%
REER	No	No
	Results	
	Test14	**Test15**
Effective growth	4.4%	4.85%
U (2050)	16%	3.6%
B/GDP (2050)	4.4%	2.5%
REER(2050/2012)	1.4	3.14

Figure 3.6 Final Baseline framing tests

3.6 Conclusion

We consider the test 15 as the macroeconomic framing test. The unemployment level is forced to reveal what trajectory of trade balance is compatible with the energy system evolution and the average wage progressing as fast as productivity (per the wage curve). So we use a closure rule which is not common in usual CGE analysis[1]. We force the unemployment level to an exogenous level. This is because the long term impact of the transition pathway on the trade balance was not easily settled in the tests discussed above, so by targeting one unemployment rate we are in fact forcing a specific level of relative domestic versus import prices (per the wage curve: that level that sets the CPI at the level that allows the real wage to grow as fast as labour productivity), which in turn induces a specific trade balance via import and export price-elasticities. The policy interpretation is that the GoI will adjust the Indian exchange rate to stabilize unemployment via monetary policy[2]. In more simple terms, the central bank in India (Reserve Bank of India) manipulates the interest rate to control inflation and unemployment. However, in the money-neutral framework of IMACLIM interest rate variations translate into adjustments of the real effective exchange rate and hence the connection of the unemployment to the trade balance which allows us to substitute the constraint $u = u_o$ to $\frac{B}{GDP} = \frac{B_0}{GDP_0}$. So the way to control the exchange rate is via the interest rate. In other words, control of unemployment rate via the interest rate is driving the exchange rate dynamics to a large extent. IMACLIM simplifies these mechanics (obviously) because it is using the wage curve which implicitly connects unemployment to the exchange rate via the CPI.

The test cases executed for KLEM model provide useful inputs for RQ 2 and 3. From our results of chapter 3, test no. 15 turned out to be the macroeconomic framing test.

[1] Standard CGE applications rather assume total wage flexibility to reach full employment, under one specific exogenous trade balance objective---commonly, an exact equilibrium, which is the condition to self-sufficiency on the domestic savings and investment market. The drawback of such model closure is the absence of control of the compared evolutions of the real wage and labour productivity.

[2] Central banks in developed countries have main mandate to control inflation i.e. to stabilize the REER in the KLEM framework. In European countries there are increasing voices in favour of adding unemployment control, if not switching to that objective only.

Chapter 3
Methodology

IMACLIM-IND model

4.1 Introduction

IMACLIM modelling platform has been developed at CIRED, International Centre for Research in

Environment and Development located in Paris since 1990's. The objective has been to articulate energy and climate policies based on the hybrid energy-economy architecture. Several versions have been developed like the global recursive dynamic model IMACLIM-R World and national versions like the recursive dynamic IMACLIM-R France and the comparative static version IMACLIM-S. Further, national versions for developing countries like Brazil and South Africa have also been developed. On similar lines, we construct the version for India IMACLIM-IND discussed in the sections below. IMACLIM-IND is a computable general equilibrium model (CGEM) designed to assess the medium- to long-term macroeconomic impacts of aggregate price or quantity-based carbon policies, in an accounting framework where economic and physical flows (with a special focus on energy balances) are equilibrated. IMACLIM models depart from the standard neoclassical CGEM in the main feature that their description of the consumers' and producers' trade-offs, and the underlying technical systems, are specifically designed to facilitate calibration on bottom-up expertise in the energy field, with a view to guaranteeing technical realism to their simulations of even large mutations of the energy systems.

Detailed energy/economy outlooks will be framed by adapting IMACLIM top-down model to the sectors that will eventually be modelled for India (see model description and equations in following subsections). This step is required because, contrary to the standard approach, IMACLIM does not postulate uniform behavioural specifications in all sectors. One important discrimination is between energy and non-energy sectors: while the latter are usually modelled as standard 'nested' trade-off structures, the former are modelled under constraint of energy systems expertise external to the model, in order to control the interface between economic and technical systems—one paramount dimension of any mitigation cost assessment (Hourcade et al., 2006). In this particular regard, the AIM/Enduse model will be coupled to IMACLIM via the iterative exchange of modelling outputs of one model as inputs of the other model, up to convergence of the set of shared variables, including, importantly, the investment requirements of the projected energy systems dynamics. This method warrants that any economic outlook produced by IMACLIM is backed by one fully consistent, detailed energy systems outlook of AIM/Enduse. The resulting 'hybrid' architecture thus combines the strengths of bottom-up and top-down energy modelling (Ghersi et al., 2006, Hourcade et al., 2006, Kriegler et al., 2015).

The salient feature of our methodology is coupling to BU based on original hybrid data reconciling national accounting and energy data. Beyond that and the corresponding accounting consistencies, all macroeconomic specifications of the model are subject to revision depending on country specifics. Concerning the growth engine, we base IMACLIM-IND on an exogenous growth engine determined by Harrod-neutral technical change with labour productivity gains. IMACLIM-IND model differs from

standard CGE model in several aspects. First is the implicit representation of capital accumulation rather than explicit recursive capital-investment dynamics, which is a simple link of proportionality between total capital stock and investment flows. Model computes the rental price of capital such that the capital market is cleared implying that the capital demand matches the exact level of capital supply, considering substitution possibilities to labour in value added. Since we are doing medium to long-term analysis in single time steps, explicit capital stock dynamics are not much relevant. Second is the use of iterative coupling to AIM/Enduse. The standard CGE nested CES production structure is not used for even the non-energy sectors. Rather, we use Leontief production function for KLE and material substitution since IMACLIM-IND embodies bottom-up information on capital intensity of some nonenergy sectors like cement, aluminium, iron and steel, and textile through exogenous capital intensities. Thus, we chose to rely more on the engineering notion derived from AIM. Third, the model includes the representation of imperfect labour market that is involuntary unemployment using an aggregate wage curve (Blanchflower and Oswald, 1995). Fourth, it represents imperfect markets by considering non-zero profits through mark-up pricing. Fifth, it breaks down capital income and Gross Fixed Capital Formation (GFCF) across three institutional agents: households, administrations and firms. Sixth, the model embarks exogenous drivers of structural change like investment trends in energy and non-energy goods, trade information on energy goods, energy consumption by households, energy goods and non-energy goods production processes.

Finally, the closure rule of the model is that of Johansen (1960), i.e. it exogenously postulates some investment effort meant to reflect trends observed in other developing and developed economies, i.e. a decrease from the uncommonly high level of 29% of GDP to a level of 27.7% in 2030 and 26% in 2050. As a consequence, households' savings adjust to the investment constraint, considering what level of foreign savings the above trade specification induces, but also the budget surpluses or deficits of both public administrations and firms. Similar to Johansen, our assumption is that of public policies incentivizing a sustainable investment effort, rather than letting possible misalignments of the domestic savings rate compromise growth. IMACLIM-IND is set-up to project the Indian economy for the medium term (2030) and long term (2050). The growth engine is based on following drivers:

- Total population and labour endowment growth are exogenous inputs in the model.

- Implicit capital accumulation computed through the proportional link between fixed capital consumption (CFC) and the total investment determined via exogenous parameter.

Calibration is the first step in model set-up where the parameters and variables are inverted for computing the output from model equations. Calibration is based on hybrid Input-Output table based on 22 sectors (Table 4.1) which involves heavy data treatment (see Chapter 2). The remainder of this Chapter provides an exhaustive description of the model's equations, while Appendix A provides the list of parameters (exogenous and calibrated) used in the model and Appendix B gives the hybrid IO table for India derived using the methodology described in Chapter 2 and the table of secondary income flows bridging the gap between primary factor income and net lending or borrowing position of four economic agents that is households, firms, government and rest of the world.

Table 4.1 IMACLIM-IND sectoral breakdown

Energy Sectors (8)		Non- energy sectors (14)	
COA	Coal	I&S	Iron & Steel
COK	Coke	CHE	Chemical
ONT	Crude oil and non-transport fuels	ALU	Aluminium
TRF	Transport fuels	CEM	Cement
BIO	Biomass	CON	Construction
GAS	Natural gas	TEX	Textile
ELE	Electricity	IND	Residual industries
REN	Renewable Energy	AGR	Agriculture
		ATR	Air transport
		WTR	Water transport
		ROA	Road transport
		RAI	Rail transport
		HOU	Housing
		SER	Other services

4.2 Formulary of IMACLIM-IND

IMACLIM-IND operates in a comparative-static framework as a system of simultaneous non-linear equations:

$$\begin{cases} f_1(x_1, x_2 \dots, x_n, y_1, y_2 \dots, y_m) = 0 \\ f_2(x_1, x_2 \dots, x_n, y_1, y_2 \dots, y_m) = 0 \\ \dots \\ f_n(x_1, x_2 \dots, x_n, y_1, y_2 \dots, y_m) = 0 \end{cases}$$

With x_i a set of n (886) variables, y_i a set of m (1874) parameters and f_i a set of n (886) functions, for

some of them linear, for some of them non-linear, in x_i. The values of some variables at calibration year constitute a specific subset of parameters, which we systematically note as the variables with a 0 index. The f_i functions embody constraints of either an accounting nature or a behavioral nature. The accounting constraints impose themselves on the modeler for the sake of consistency. The behavioral constraints, quite distinctively, convey the modeler's views on economic causalities and correlations.

4.2.1 Price system

IMACLIM-IND model differs from the standard CGE model in terms of price and income structural constraints by accounting for non-zero profits via mark-up pricing, considering specific margins for energy goods and by considering sector-specific wages based on satellite accounts of full-time equivalent employment by sector.

4.2.1.1 *Primary factor prices*

IMACLIM-IND calibrates the labour inputs to productions on employment statistics measured in fulltime equivalents. This induces sector-specific average net wages w_i. The model considers that all w_i record the same variation δ_w from their base-year levels w_{i0}:

$$w_i = (1 + \delta_w)\, w_{i0} \tag{34}$$

The endogenous adjustment of δ_w follows a 'wage curve' constraint to the labour market clearing (see section 4.2.6.2). In each sector, labour costs p_{Li} are the sum of net wages w_i and of labour taxes at τ_{LTi} rate, which are the social contributions of employers and employees—provident funds in the case of India. The labour tax rates τ_{LTi} are parameters calibrated at base year. Insurance contribution by agent
implies the payroll taxes or contribution to provident fund. The data source for this was the employees' provident fund organization annual report, 2012-13. We consider only the salaried class who contributes to provident fund and not the other status occupations. The informal sector, which occupies the unskilled jobs mostly, does not contribute to provident fund (GoI, 2015).

$p_{Li} = (1 + \tau_{LTi})\, w_i$ (35) The average wage across all sectors w is:

$$w = \frac{\sum_{i=1}^{n} w_i\, \lambda_i\, Y_i}{\sum_{i=1}^{n} \lambda_i\, Y_i} \tag{36}$$

The cost of capital write-offs p_K is common to all sectors. At base year, its normalisation at 1000 allows calibrating volumes of fixed capital consumption for all sectors based on the money-metric fixed capital consumption of our hybrid IOT. At projection horizon, it is an endogenous variable of IMACLIM-IND, which adjusts to clear the capital write-off capacity (see section 4.2.6.4).

4.2.1.2 *Output prices*

The producer price of good i, p_{Yi}, proceeds from the sum of input costs, output taxes at a τ_{Yi} rate, and a mark-up rate π_i corresponding to the net operating surplus.

$$p_{Yi} = \sum_{j=1}^{n} p_{ji}\, \alpha_{ji} + p_{Li}\, \lambda_i + p_K\, \kappa_i + \tau_{Yi}\, p_{Yi} + \pi_i\, p_{Yi} \tag{37}$$

Input costs are the products of input prices and input intensities or technical coefficients λ_i (labour intensity), κ_i (fixed-capital-consumption intensity) and α_{ji} (intensities in secondary factors including energy intensities). The mark-up rate π_i is calibrated at base year and held constant in further years.

A Fisher index (geometric mean of the Laspeyres and Paasche indexes) measure the aggregate evolution of output prices:

$$OPI = \sqrt{\frac{\sum_i p_{Yi}\, Y_{i0}\ \sum_i p_{Yi}\, Y_i}{\sum_i p_{Yi0}\, Y_{i0}\ \sum_i p_{Yi0}\, Y_i}} \tag{38}$$

4.2.1.3 *Import and average supply prices*

Import prices p_{Mi} are exogenous. 'Other Services' imports act as *numéraire* of IMACLIM-IND and their price is therefore maintained at base-year value. Energy import prices variations from base year reflect AIM/Enduse assumptions.

We consider the imported and domestic energy goods to be homogenous rather than assuming Armington specification as commonly done in CGE models. This is to maintain an explicit account of physical volumes, which is one primary component of our approach of modelling.[1] For the sake of simplicity, we treat non-energy goods similarly to energy goods. The average supply price of good i, p_{Si}, is therefore the weighted average of the output and import prices of good i, with the output and import volumes as weights.

$$p_{Si} = \frac{p_{Yi}\, Y_i + p_{Mi}\, M_i}{Y_i + M_i} \tag{39}$$

4.2.1.4 *Intermediate consumption prices*

The price of intermediate goods p_{ij}, i.e. the purchaser's price of good i for producing good j, is equal to the supply price augmented from trade τ_{CMi} and transport τ_{TMi} margins, specific margins τ_{SMij} and excise taxes:

[1] The Armington specification differentiates the domestic and imported versions of goods. It creates a hybrid of both goods, whose volume does not proceed from the summing of those of the imported and domestic goods.

$$p_{ij} = p_{Si}\,(1 + \tau_{CMi} + \tau_{TMi} + \tau_{SMij}) + t_{ETij} \tag{40}$$

Specific margins embody deviations of purchaser prices from the average output price augmented by the relevant taxes and trade and transport margins. In IMACLIM-IND, where the process of data hybridisation only regards energy goods, they are nil for non-energy goods. For all goods and at all modelled horizons, we keep them at their calibration values.

We also keep all positive trade and transport margins at their calibration values at all modelled horizons. The negative margins, which correspond to those sectors providing the underlying trade and transport services, adjust to warrant the accounting balances:

$$\sum_i \tau_{CMi}\, p_{Si}\, \left(\sum_{j=1}^{n} \alpha_{ij}\, Y_j + C_i + G_i + I_i + X_i\right) = 0 \tag{41}$$

and

$$\sum_i \tau_{TMi}\, p_{Si}\, \left(\sum_{j=1}^{n} \alpha_{ij}\, Y_j + C_i + G_i + I_i + X_i\right) = 0 \tag{42}$$

In the case of transport services, which flow from four different sectors, we assume that the four corresponding negative margins adjust according to one single endogenous variation δ_{TM}:

$$\forall i \in \{ATR, WTR, ROA, RAI\}\ \tau_{TMi} = (1 + \delta_{TM})\ \tau_{TMi0} \tag{43}$$

The lack of statistics prevents differentiating either τ_{TMi} or τ_{CMi} by end-use.

4.2.1.5 *Final consumption prices*

The purchaser's prices of good i for households p_{Ci}, government p_{Gi}, investment p_{Ii} and foreign agents (export price) p_{Xi} follow definitions similar to that of intermediate prices, with the difference of the additional value-added tax at rate τ_{VATi} for all sales but exports, which do not either bear any excise tax.

$$p_{Ci} = (p_{Si}\,(1 + \tau_{CMi} + \tau_{TMi} + \tau_{SMCi}) + t_{ETCi})\,(1 + \tau_{VATi}) \tag{44}$$

$$p_{Gi} = (p_{Si}\,(1 + \tau_{CMi} + \tau_{TMi} + \tau_{SMGi}) + t_{ETGi})\,(1 + \tau_{VATi}) \tag{45}$$

$$p_{Ii} = (p_{Si}\,(1 + \tau_{CMi} + \tau_{TMi} + \tau_{SMIi}) + t_{ETIi})\,(1 + \tau_{VATi}) \tag{46}$$

$$p_{X_i} = p_{S_i}\,(1 + \tau_{CM_i} + \tau_{TM_i} + \tau_{SMX_i}) \qquad (47)$$

Similar to the output price index OPI, a Fisher consumer price index (CPI) measures the aggregate evolution of consumer prices:

$$CPI = \sqrt{\frac{\sum_i p_{c_i}\, c_{i_0}\ \sum_i p_{c_i}\, c_i}{\sum_i p_{c_{i0}}\, c_{i_0}\ \sum_i p_{c_{i0}}\, c_i}} \qquad (48)$$

4.2.2 Households

4.2.2.1 *Income, savings and investment*

The after-tax disposable income of households R_H proceeds from primary factor incomes, social transfers, property income and an aggregate of other secondary transfers.

$$R_H = \sum_i w_i\, l_i\, Y_i + \omega_{KH}\, GOS + \sum_{i=P,U,T} \rho_i\, N_i + \omega_{OTH}\, GDP - i_H\, D_H - \tau_{IT}\, R_H - t_H\, CPI\, N_T \qquad (49)$$

$\sum_i w_i\, l_i\, Y_i$ is the total net income from labour i.e. the sum of wages obtained from all economic sectors.

We obtained labour disaggregation across sectors from the NSS employment survey, except for the 'Renewables' sector, which we obtained from the IRENA (2017) report on 2013 renewable jobs, and for the 'Housing' sector, which we disaggregated from services by assuming homogeneous average wage. ω_{KH} GOS is the share ω_{KH} of total capital income, or gross operating surplus GOS, directly accruing to households in the form of, mainly, housing rents (imputed or real). Social transfers involve pensions $\rho_P\ N_P$, unemployment transfers $\rho_U\, N_U$ and other social transfers $\rho_T\, N_T$ like support in the form of minimum income given by the government. ρ_i stands for per capita transfers and N_i for a target population: exogenous pensioned population N_P, endogenous unemployed population N_U or exogenous total population N_T. Other transfers form a constant ω_{OTH} share of GDP calibrated at base year. They include marginal income sources like lottery gains or remittances. Property income is the interest payment on the net debt D_H at rate i_H resulting from the balance of income from financial assets, bonds or credit, and interest payments on liabilities. Income taxes are paid at rate τ_{IT} on disposable income R_H. A per capita t_H amount of other direct taxes (housing taxes, land taxes, etc.) is paid as well, which evolves as the consumer price index CPI.

Households savings at rate τ_S adjust to balance investments and savings (see section 4.2.6.4). The consumption budget of households is equal to the disposable income net of savings:

$R_C = (1 - \tau_S)\, R_H$ (50) The investment effort of households $GFCF_H$ is in constant proportion to their disposable income R_H. We computed the contributions of agents to total gross fixed capital formation

using the information on key economic indicators for institutional sectors from Ministry of Statistics and Programme Implementation (MOSPI) (see Appendix B, Table B2, Page 191).

$$\frac{GFCF_H}{R_H} = \frac{GFCF_{H0}}{R_{H0}} \tag{51}$$

The net lending or borrowing (NLB) of households NLB_H is the difference between their disposable income and their consumption and investment.

$$NLB_H = R_H - R_C - GFCF_H \tag{52}$$

4.2.2.2 *Consumption choices*

Households' consumptions of energy goods proceed from the coupled AIM/Enduse bottom-up model. The iterative coupling procedure warrants that they are consistent with the evolution of households' income. Non-energy goods are in two sets. One set resorts to elasticities of per capita consumption $\frac{C_i}{N_T}$ to real per capita consumption budget:

$$\forall i \in \{TEX, IND, AGR, ATR, RAI, ROA, HOU\}\ C_i = C_{i0}\ \frac{N_T}{N_{T0}} \left(\frac{R_C}{N_T\ OPI} \frac{N_{T0}}{R_{C0}}\right)^{\sigma_{CRi}} \tag{53}$$

For numerical convenience, we use the output price index OPI as a proxy of the CPI deflator to compute real variations of the consumption budget at current prices R_C.

For lack of better assumption, the remaining set of non-energy goods mobilises constant shares of the consumption budget remainder:

$$\forall i \in A = \{I\&S, CEM, CHE, ALU, CON, WTR, SER\}\ \frac{p_{Ci}\ C_i}{R_C - \sum_{j \notin A} p_{Cj}\ C_j} = \frac{p_{Ci0}\ C_{i0}}{R_{C0} - \sum_{j \notin A} p_{Cj0}\ C_{j0}} \tag{54}$$

The residual budget shares of I&S, ALU, CON and WTR are in fact nil. Cement (CEM) mobilises a mere 0.1% and chemical and petrochemical products (CHE) 5%. The bulk of the residual budget therefore accrues to services (SER).

4.2.3 Firms

4.2.3.1 *Income and investment*

Similarly to households, firms earn a share ω_{KF} of capital payments GOS and a share ω_{OTF} of GDP as other transfers. We calibrate both shares at base year and keep them constant in projections. Firms pay interests at rate i_F on their net debt D_F as well as corporate taxes at rate τ_{CT} on their gross operating surplus GOS.

$$R_F = \omega_{KF}\ GOS + \omega_{OTF}\ GDP - i_F\ D_F - \tau_{CT} GOS \tag{55}$$

Their investment effort is equal to total investment net of the investment of households and public administrations:

$$GFCF_F = \sum_i p_{Ii}\, I_i - GFCF_H - GFCF_G \tag{56}$$

The fact that total investment is a fixed share of GDP at current prices (see Equation 81) and the investment behaviours of households and public administrations warrants a relatively stable investment effort by firms.

Similar to households as well, the net lending or borrowing of firms is then:

$$NLB_F = R_F - GFCF_F \tag{57}$$

4.2.3.2 *Production trade-offs*

Coupling with the AIM/Enduse model provides information on the evolution of the energy intensities of all sectors as well as on the capital intensity of a subset of them: all 8 energy sectors, heavy industries I&S, CHE, CEM, ALU and textile TEX. IMACLIM-IND builds on such information, which it treats as exogenous Leontief coefficients. Additionally, it keeps constant at calibration value all technical coefficients that cannot be inferred from AIM (typically non-energy intensities). The coupling process via iteration to convergence warrants that AIM/Enduse energy consumptions and annualised investment costs are consistent with IMACLIM-IND activity drivers.

For the 9 non-energy sectors whose capital intensity is not traced by AIM (CON, IND, AGR, ATR, WTR, ROA, RAI, HOU, SER), IMACLIM-IND considers CES substitutability between capital K and labour L, to form a value-added aggregate KL: $KL_i = (a_{KLi}\, K_i^{\rho_{KLi}} + b_{KLi}\, (\phi_L L_i)^{\rho_{KLi}})^{\frac{1}{\rho_{KLi}}}$, taking account of labour productivity gains ϕ_L. The ρ_{KLi} parameter is in fact a function of σ_{KLi} the elasticity of substitution of K to L in sector i: $\rho_{KLi} = \frac{\sigma_{KLi}-1}{\sigma_{KLi}}$. The a_{KLi} and b_{KLi} paramaters are calibrated at base year, they relate to the relative cost shares of K and L for the different activity sectors. At projection year, facing prices p_K and p_{Li}, cost minimization yields for sectors $i \in \{CON, IND, AGR, ATR, WTR, ROA, RAI, HOU, SER\}$:

$$L_i = \frac{1}{\phi_L}\left(\frac{b_{KLi}\,\phi_L}{p_{Li}}\right)^{\sigma_{KLi}} \left(a_{KLi}{}^{\sigma_{KLi}}\, p_K{}^{1-\sigma_{KLi}} + b_{KLi}{}^{\sigma_{KLi}} \left(\frac{p_{Li}}{\phi_L}\right)^{1-\sigma_{KLi}}\right)^{\frac{-1}{\rho_{KLi}}} KL_i \tag{58}$$

$$K_i = \left(\frac{a_{KLi}}{p_K}\right)^{\sigma_{KLi}} \left(a_{KLi}{}^{\sigma_{KLi}}\, p_K{}^{1-\sigma_{KLi}} + b_{KLi}{}^{\sigma_{KLi}} \left(\frac{p_{Li}}{\phi_L}\right)^{1-\sigma_{KLi}}\right)^{\frac{-1}{\rho_{KLi}}} KL_i \tag{59}$$

All secondary factor intensities are exogenous, either taken from AIM (energy intensities) or constant at calibration year value (non-energy intensities). The KL intensity of concerned productions is constant (Leontief assumption):

$$\forall i \in \{CON, IND, AGR, ATR, WTR, ROA, RAI, HOU, SER\} \ \frac{KL_i}{Y_i} = \frac{KL_{i0}}{Y_{i0}} \tag{60}$$

4.2.3.3 *Gross operating surplus*

Gross operating surplus accruing from productive activities is the sum of capital write-offs, profits at mark-up rates π_i and specific margins SM, which are in fact cost differentials between end-uses of each energy sector. From the literature on national statistics, we assume that GOS divides amongst the three agents, corporation, government and households, in ratios of 67%, 14% and 19% respectively.

$$GOS = \sum_{i=1}^{n} \left(p_K \, \kappa_i \, Y_i + \pi_i \, p_{Y_i} \, Y_i\right) + SM \tag{61}$$

Total specific margins are:

$$SM = \sum_i \left(\sum_j \tau_{SMICi} \, p_i \, \alpha_{ij} \, Y_j + \tau_{SMCi} \, p_i \, C_i + \tau_{SMGi} \, p_i \, G_i + \tau_{SMXi} p_i \, X_i\right) \tag{62}$$

4.2.4 Public administrations

4.2.4.1 *Public income*

The gross disposable income of public administrations R_G derives from taxes and social security contributions T, exogenous ω_{KG} and ω_{OTG} shares of the GOS (that of public companies) and GDP, corrected from transfers to households $\sum_j \rho_j \, N_j$ and interest payments at rate i_G on the net public debt D_G.

$$R_G = T + \omega_{KG} \, GOS + \omega_{OTG} \, GDP - \sum_{i=U,P,T} \rho_i \, N_i - i_G \, D_G \tag{63}$$

Tax revenue T comprises primary factor and output taxes, the VAT and excise taxes, the income tax and other direct taxes and the corporate tax.

$$T = \sum_{i=1}^{n} \tau_{LTi} \, w_i \, l_i \, Y_i + \tau_{Yi} \, p_{Yi} \, Y_i + \frac{\tau_{VAT_i}}{1+\tau_{VAT_i}} \left(p_{C_i} \, C_i + p_{G_i} \, G_i + p_{I_i} \, I_i\right) + \sum_i \sum_j t_{ETij} \, \alpha_{ij} \, Y_j$$

$$+ t_{ETCi} \, C_i + t_{ETGi} \, G_i + t_{ETIi} \, I_i + \tau_{IT} \, R_H + t_H \, CPI \, N_T + \tau_{CT} \sum_i \pi_i \, p_{Yi} \, Y_i \tag{64}$$

Per capita social transfers are in constant proportion to the average wage:

$$\forall i \in \{U, P, T\} \ \frac{\rho_i}{w} = \frac{\rho_{i0}}{w_0} \tag{65}$$

4.2.4.2 *Public expenditures and budget balance*

Public expenditures $\sum_i p_{Gi}\, G_i$ and investment $GFCF_G$ are both constant shares of GDP at current prices:

$$\frac{\sum_{i=1}^{n} p_{Gi}\, G_i}{GDP} = \frac{\sum_{i=1}^{n} p_{Gi0}\, G_{i0}}{GDP_0} \tag{66}$$

$$\frac{GFCF_G}{GFCF_{G0}} = \frac{GDP}{GDP_0} \tag{67}$$

Public expenditures distribute across sectors by assuming homothetical variations from calibration year levels:

$$G_i = \beta_G\, G_{i0} \tag{68}$$

The public budget balance or net lending or borrowing of public administrations is:

$$NLB_G = R_G - \sum_i p_{Gi}\, G_i - GFCF_G \tag{69}$$

4.2.5 Rest of the world

4.2.5.1 *International trade*

International trade of energy goods is exogenous from the AIM/Enduse model. For non-energy goods, the competition on international markets depends on the relative prices of goods. The contribution of trade to resources and uses i.e. the ratios of imports M_i and exports X_i to total supply $Y_i + M_i$ are elastic to the evolution of domestic (p_{Yi} or p_{Xi}) versus foreign (p_{Mi}) prices.

$$\frac{M_i}{Y_i + M_i} = \frac{M_{i_0}}{Y_{i_0} + M_{i_0}} \left(\frac{p_{M_0}\, p_{Y_i}}{p_{Y_0}\, p_{M_i}} \right)^{\sigma_{M p_i}} \tag{70}$$

$$\frac{X_i}{Y_i + M_i} = \frac{X_{i_0}}{Y_{i_0} + M_{i_0}} \left(\frac{p_{X_0}\, p_{M_i}}{p_{M_0}\, p_{X_i}} \right)^{\sigma_{X p_i}}$$ (71) The ratio between the domestic and international vectors of prices does not result from a trade balance assumption as in standard CGE models. Rather, it is dictated by the assumption of an unemployment rate stabilised at base year level:

$$u = u_0 \tag{72}$$

The purpose of not fixing the trade balance is to allow assessing the transitory impact of low-carbon pathways on trade deficits. Stabilising the unemployment rate requires setting the ratio between domestic and international prices at some specific level, under constraint of the above import and export

specifications. The policy interpretation is that of the Indian government manipulating the nominal exchange rate with that objective as target.

4.2.5.2 *Capital flows*

The Rest of the world (ROW) agent balances out trade (by selling imports $\sum_i p_{Mi} M_i$ and buying exports $\sum_i p_{Xi} X_i$), property income and interest payments. Its net lending or borrowing capacity NLB_{ROW} is thus:

$$NLB_{ROW} = \sum_i p_{Mi} M_i - \sum_i p_{Xi} X_i - \sum_{j=H,F,G} i_j D_j - \sum_{j=H,F,G} \omega_{OTj} GDP \quad (73)$$

4.2.6 Market clearings

4.2.6.1 *Goods markets*

The balance of goods markets is between resources, which comprise domestic production Y_i and imports M_i, and uses, which consists of households' and public consumptions C_i and G_i, immobilisations I_i and exports X_i. For energy goods, the data hybridization process results in this equation being expressed in million tons-of-oil-equivalent (Mtoe), inconsistency with the 2012 Indian energy balance of the International Energy Agency (IEA). The public consumptions and immobilisations of all energy goods are nil, by national accounting convention for the former and by definition for the latter.

$$Y_i + M_i = \sum_{j=1}^{n} \alpha_{ij} Y_j + C_i + G_i + I_i + X_i \quad (74)$$

4.2.6.2 *Labour market*

Total labour demand from sector i is:

$$L_i = \lambda_i Y_i \quad (75)$$

Labour market clearing requires that labour demand and unemployment balance out labour supply from households. One of the key structural assumptions of IMACLIM-IND is indeed to consider equilibrium unemployment u. The labour supply of households or labour endowment L is an exogenous parameter computed from International Labour Organisation (ILO) 2018 estimates until 2021 and considering the 2020 ratio of labour to working age (20 to 69) population (UN, 2015) from 2021 on.

$$\sum_{i=1}^{22} L_i = (1-u) L \quad (76)$$

The unemployed population N_U is:

$$N_U = u\,L \tag{77}$$

A wage curve describes the correlation between the unemployment rate and the average wage

(Blanchflower and Oswald, 2005). The purchasing power of wages—the average wage at current prices w deflated by the consumer price index CPI—is elastic to unemployment variations $\frac{u}{u_0}$ with σ_{wu} elasticity, around a trend defined by labour productivity gains ϕ_L:

$\frac{w}{CPI} = \phi_L\, w_0 \left(\frac{u}{u_0}\right)^{\sigma_{wu}}$ (78) Considering current implementations in single time steps from 2012 to 2030 and 2050, IMACLIM-IND simplifies capital accumulation by assuming that total fixed capital

4.2.6.3 *Capital markets*

Total capital write-offs from sector i are:

$$K_i = \kappa_i\, Y_i \tag{79}$$

consumption grows proportionally to investment flows:

$$\sum_{i=1}^{22} K_i = \beta_I \sum_{i=1}^{22} K_{i0} \tag{80}$$

The cost of capital write-offs p_K adjusts to constrain them to this target level (see section 4.2.1.1).

4.2.6.4 *Investment and savings*

Rather than considering fixed domestic and foreign savings and closing on investment (neoclassical closure rule of the standard CGE model), IMACLIM-IND considers a fixed investment effort ('Johansen closure' following Sen, 1963), flexible foreign savings—induced by the exchange rate adjustments allowing to maintain the unemployment level (see Equation 72)—and closes on domestic savings by adjustment of households' savings rate τ_S. Investment is an exogenous GDP share:

$$\sum_{i=1}^{22} p_{Ii}\, I_i = s_I\, GDP \tag{81}$$

The investment supply of households is a fixed share of their income and that of public administrations a fixed share of GDP. The investment supply of firms adjusts to match investment demand (see Equation 56 above).

Similar to public expenditures, the investment effort distributes across sectors by assuming homothetical variations from calibration values i.e. constant composition of the homogenous (across sectors) investment good:

$$I_i = \beta_I \, I_{i0} \tag{82}$$

4.2.7 GDP

GDP is the sum of final expenditures:

$$GDP = \sum_{i=1}^{22} p_{C_i} C_i + p_{G_i} G_i + p_{I_i} \, I_i + p_{X_i} \, X_i - p_{M_i} \, M_i \tag{83}$$

4.2.8 Net debts of agents

Depending on its sign at calibration year and at the projection horizon, the NLB of agents accumulates into debt or credit. The assumption is that agents see their NLB net of interest payments or Primary Balance evolve linearly over the y years between the calibration year and the projection year. Interests accrue at agent-specific rate i_j, $j \in \{H, F, G\}$, while monetary inflation at rate τ_{IM} erodes the real value of the calibration-year net debt D_{j0}. By recursion and using known sums of terms of geometric suites, the net debt at projection horizon is therefore, for $j \in \{H, F, G\}$:

$$D_j = \left(\frac{1+i_j}{1+\tau_{IM}}\right)^y \, D_{j0} - A_{Dj} \left(PB_{j0} - (y-1)\frac{PB_j - PB_{j0}}{y}\right) + B_{Dj} \frac{PB_j - PB_{j0}}{y} \tag{84}$$

with

$$PB_j = NLB_j + i_j \, D_j \tag{85}$$

$$A_{Dj} = \frac{1 - \left({1+i_j}/{1+\tau_{IM}}\right)^y}{1 - {1+i_j}/{1+\tau_{IM}}} \tag{86}$$

$$B_{Dj} = \frac{(1+\tau_{IM})^y (1+i_j) - y\,(1+\tau_{IM})\,(1+i_j)^y + (y-1)(1+i_j)^{y+1}}{(\tau_{IM}+i_j)^2\,(1+\tau_{IM})^{y-1}} \tag{87}$$

The net debt of foreign agents D_{ROW} balances out domestic assets and liabilities:

$$D_{ROW} = -\sum_{j=H,F,G} D_j \tag{88}$$

4.3 Sensitivity Analysis

We conduct sensitivity analysis of our key macroeconomic results (BAU[1]) to check their variations with changes in exogenous parameters shaping the labour market and foreign trade flows. First, we test for variation in price elasticity for exports. Results show that there is not much variation in real GDP but the trade balance contribution to GDP changes significantly. Varying the price elasticity of imports impacts the trade balance ratio, household consumption and the CPI. We find that our results are sensitive to trade elasticities. Lastly, with respect to income elasticities, the macroeconomic results are not impacted by even large variation implying that the model results are insensitive to income elasticities. Our results are not sensitive to real wage elasticity to unemployment since our analysis is based on the specific rule of keeping the unemployment rate constant.

4.4 Convergence of IMACLIM-IND and AIM/Enduse model

IMACLIM-IND is based on static comparative framework involving simultaneous solving of non-linear equations with little optimization, while the bottom-up AIM-Enduse model involves optimization using linear programming with large number of constraints. The existence of a convergence point is guaranteed by the effort put towards bringing the two models into consistency in terms of the data inputs and model assumptions. As the two models are consistent from the very start of setup, it is unlikely that the two will diverge. The convergence point is likely to be unique because one model is based on linear programming which always gives one unique solution. So post-convergence, we expect to arrive at one unique solution.

[1] BAU scenario assumptions are discussed in chapter 5

Chapter-4
Research Question 2

Achieving sustainable development in India along the low carbon pathways: Macroeconomic assessment

5.1 Introduction

Developing countries like India face multiple challenges echoing the key dimensions of the Sustainable Development Goals (SDGs) of the United Nations. It thus counts around 269 million people living in poverty (Planning Commission, 2012), approximately 500 million deprived of clean cooking fuel, 304 million having no access to electricity (NEP, 2017), 163 million people without access to safe drinking water (WaterAid, 2018), close to 1.7 million people homeless (Census, 2011) and 48% of rural households lacking basic socio-economic services (SECC, 2015). Moreover, post-2020 climate actions outlined in India's Nationally Determined Contribution (NDC) under the Paris agreement envision development along low-carbon emission pathways. Therefore, the enormous developmental needs have to be balanced with the emission reduction targets. Currently, nearly three fourth of electricity generated in India is obtained from coal, and crude oil and refined petroleum products constitute about 60% of total imports. In such conditions, achieving rapid economic growth and GHG mitigation targets concurrently in Indian economy which is predominantly fuelled by fossils, can have wide ranging macroeconomic implications. Though climate action can help redress the trade unbalance, the transition to non-fossil fuels can prove to be costly.

India has been growing at a remarkable GDP growth rate of 7-8% annually since economic liberalization in 1991. The share of agriculture in GDP has gone down from 42% in 1970 to mere 17% in 2015, and continues to decline. In 2016, services and industry sectors constituted 53% and 31% of GDP respectively (Economic Survey, 2018).

India has committed to reduce the GHG emission intensity of GDP by 33-35% from 2005 levels and raising the non-fossil fuel based power generation capacity to 40% by 2030 (MoEFCC, 2015). For meeting these targets, Government of India (GoI) has taken several measures under the National Action
Plan on Climate Change (NAPCC). India has set the target of achieving 175 GW from 35 GW (as in

2015) of power capacity through renewable sources by 2022 including solar target of 100 GW by 2022. There is emphasis on improving energy efficiency as the government is taking several demand-side management initiatives and targeting strict norms for the energy-intensive industries. Further, GoI has taken a joint initiative with State Governments to provide 24X7 Power for All (PFA) by 2022.

5.2 Methodology

We follow the methodology described in chapter 4. The exogenous structural drivers of the model are i) energy consumption by sectors and households ii) energy trade flows iii) energy-intensity technical coefficients iv) sector-wise capital intensities, which shift both on the supply and end-use

sides. Figure 5.1 below illustrates the iteration process followed for different scenarios so that carbon output is stabilized at BAU level. After putting the scenario assumptions in IMACLIM, we get sectoral end-use demands which are fed into AIM. AIM run generates energy-related outputs, which are given as inputs to IMACLIM. We check if the carbon output from IMACLIM is stabilized at BAU level. We apply carbon caps in AIM, which further gives energy and capital intensities numbers. These numbers are fed into IMACLIM. This process is followed iteratively until carbon emissions are stabilized in 2DegMG,

2DegHG and 2DegLG scenarios at BAU level. The results from the process are discussed in Section 5.4.

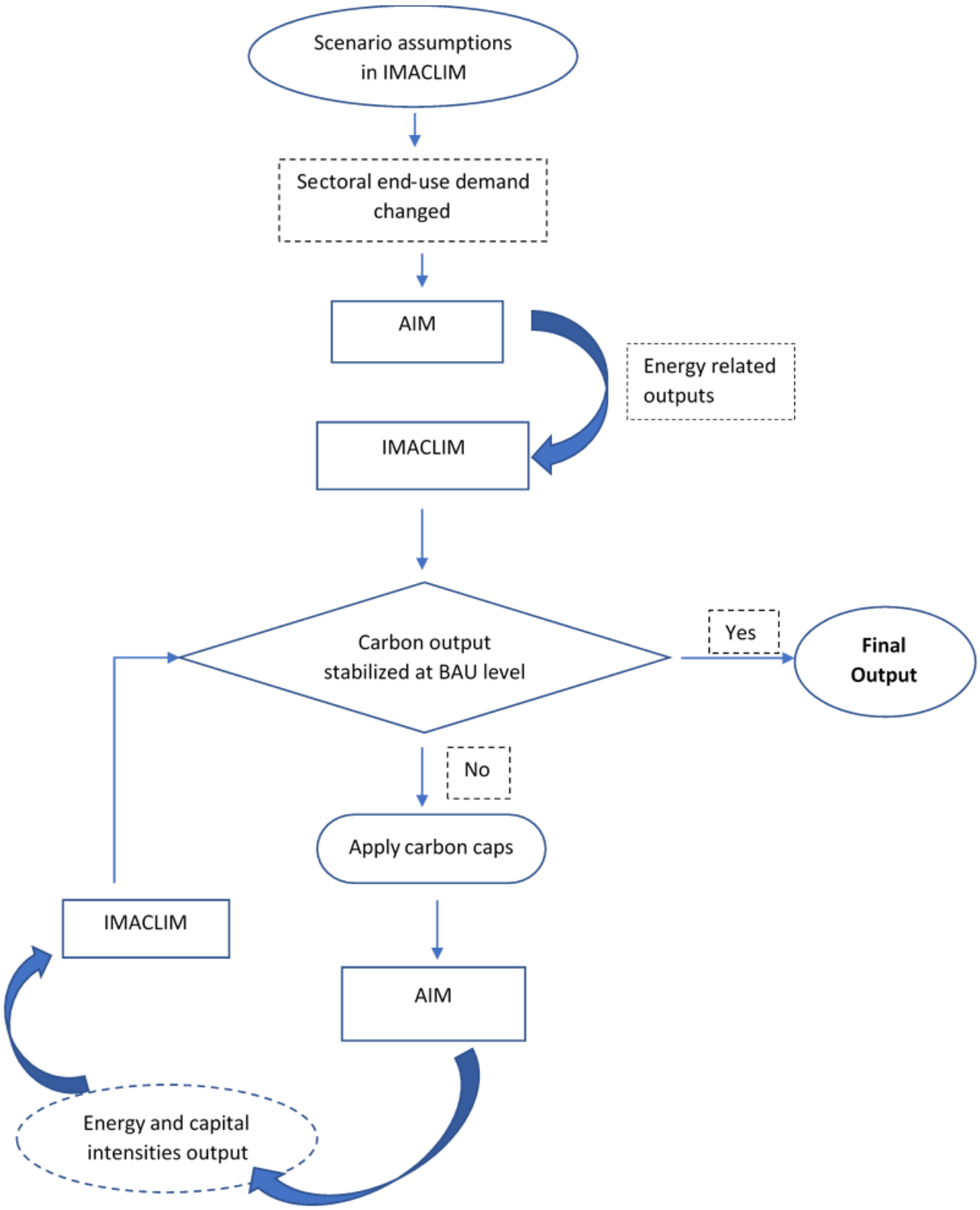

Figure 5.1 Illustration of iteration process

5.3 Scenario architecture

We model 4 scenarios BAU, 2DegMG, 2DegHG and 2DegLG and project them at 2030 and 2050 milestones. These scenarios provide support in determinating the feasibility and related trade-offs in achieving high growth in a low carbon economy. The assumptions are illustrated in Table 5.1 and discussed in the subsections below.

5.3.1 BAU scenario

BAU scenario follows the current pattern of emission and development dynamics. It builds on the recent energy and climate change policies as outlined in NAPCC (PMCoCC, 2008) and Indian INDCs submitted to UNFCCC (MoEFCC, 2015). It simulates up-to-date macroeconomic and social dynamics. We generate a consistent energy-economy representation by integrating the energy and socio-economic visions.

The strategy for mitigation includes eight National Missions in areas like solar energy, enhanced energy efficiency, sustainable habitat, water, sustaining the Himalayan ecosystem, green India, sustainable agriculture and strategic knowledge for climate change (PMCoCC, 2008), and the State Action Plans on Climate Change (SAPCC) consistent with NAPCC prepared by 32 States/Union Territories. In particular, this scenario includes the voluntary goal announced by India to reduce the emission intensity of its GDP by 33-35% from 2005 levels by 2030. Other supplementing strategies include policies like the National Electricity Policy (NEP) (2005), which focusses on universalizing the access to electricity, National Policy for Farmers (NPF) (2007) and National Mission on Sustainable Agriculture (NMSA), which underscore the sustainable development of agriculture, and Energy Conservation Act (2001) which encourages the efficient use of energy. In the rest of this subsection, we provide the BAU scenario narrative along with the main exogenous assumptions.

- We assume India's population to grow at 1.06 % annually on average from 2013-2030 and reaches 1.5 billion people by 2030 (UNPD, 2015). Structural changes in labour market yield an increase of 1.57% per year of labour endowment finally reaching 644 million workers in 2030.

  Based on expert opinion, we assume that the productivity of Indian labour grows at a fast pace with average annual growth rate of 4.9 % from 2013 until 2030 (Table 5.1). Unemployment rate is assumed to remain constant at 2.23% following the National Sample Survey Office (NSSO)
  (2013) report and Mehrotra et al. (2014). We consider a decreased rate of Gross Fixed Capital Formation (GFCF) in future since the period of infrastructure build-up is assumed to end by 2030 and investment slowly reaches lower levels where it involves more retrofitting/replacing and less new infrastructure.

- The structure of energy system and CO2 emission trajectory is mainly based on current dynamics. The exogenous parameters controlling energy systems in AIM model are

technologies availability, technology share, device lifetime, power capacity constraint, energy price, technology price, operational cost and technology efficiency while the technology transition is endogenous based on cost optimization. Coal will continue to play an important role in power sector, however, the role of renewables like solar, wind and hydro will become prominent based on capacity constraint. Also, biomass use in power generation will increase with biomass pellets being used along with coal in all thermal power plants. In case of energy-intensive manufacturing sector, energy-efficient technologies and cleaner fuels will be selected endogenously in industries like iron and steel, aluminium, cement, chemical and petrochemical, textile, paper and pulp, and construction. Transport sector will use more clean energies like ethanol-blended petrol. Energyefficient appliances like CFL and LED will replace the old inefficient technologies in commercial and residential buildings endogenously as a consequence of device improvement coefficients.

- To analyse the macroeconomic implications, we set up IMACLIM-IND to detail 22 sectors, 8 of which energy sectors (coal, coke, crude oil and non-transport fuels, transport fuels, biomass,

natural gas, electricity, renewables) and 14 non-energy sectors (iron & steel, chemical & petrochemical, aluminium, cement, construction, textile, residual industries, agriculture, air transport, water transport, road transport, rail transport, housing and other services). Table 5.1

Main assumptions of four scenarios

	BAU	2DegMG	2DegHG	2DegLG
Labour supply	+1.57% per year from 2013 to 2030 then +0.72% from 2030 to 2050			
Labour productivity	+4.9% per year from 2013 to 2030 then +4.8% from 2030 to 2050	+4.9% per year from 2013 to 2030 then +4.8% from 2030 to 2050	+5.8% per year from 2013 to 2030 then +5.8% from 2030 to 2050	+3.9% per year from 2013 to 2030 then +3.8% from 2030 to 2050
Resulting potential growth rate	+6.5% per year from 2013 to 2030, then +5.6% from 2030 to 2050	+6.5% per year from 2013 to 2030, then +5.6% from 2030 to 2050	+7.5% per year from 2013 to 2030, then +6.6% from 2030 to 2050	+5.5% per year from 2013 to 2030, then +4.6% from 2030 to 2050
Energy policy	NAPCC, INDC	Advanced INDC, CCS in power plants and energyintensive industries	More renewables, higher energy efficiency, more CCS, BECCS	NAPCC, INDC (less advanced)
Behavioural change over BAU	None	More public transport, dematerialization, waste recycling	Sustainable actions in all sectors, public & shared transport, dematerialization, demand reduction, waste recycling	None

Cumulative CO_2 emissions (Billion tons) from energy, 2011 to 2050.	165	123	123	123

5.3.2 Other scenarios

2 Degree Medium Growth (2DegMG) scenario embodies advanced contribution targets for the period 2021 to 2030 compared to those submitted under the Paris agreement. It aims at reducing the emission intensity of GDP by 40 to 45 percent by 2030 from 2005 level, achieving 50% of installed power capacity from non-fossil fuels by 2030 and improving energy efficiency in all sectors of the economy. In the power, buildings and agriculture sector, we set advanced targets so that renewables play an important role as energy source. Solar, wind, hydro and biomass power capacity will reach 150 GW, 70 GW, 25 GW and 35 GW by 2030 respectively. We additionally assume that aggregate technical and commercial losses decrease by 6-8% and new technologies like smart and micro grids are introduced by 2030. Energyintensive industries like iron and steel, aluminium, chemical and petrochemical, cement, construction and textile become more energy-efficient through implementation of Perform, Achieve and Trade (PAT) scheme[1] with enhanced coverage of sectors and plants. Public transport systems like Dedicated Freight Corridors (DFCs) and railways play a prominent role for reducing the carbon emissions in the transport sector.

Carbon budget is constrained to 165 billion tons (Bt) CO2-equivalent (CO2eq) cumulative emissions during 2011-2050 (Table 5.1). In the power sector, low efficiency coal based power plants are shut down and energy efficient technologies like super-critical pulverized coal and Integrated Gasification Combined Cycle (IGCC) power plants are picked via exogenous share constraint. This induces higher capital mobilization in power sector. Renewables capacity and gas-based generation is increased along with storage technologies. Carbon Capture and Storage (CCS) technology penetrates in power and energy-intensive industry sectors. Carbon constraint will cause some policy amendments like PAT will now also cover the Micro, Small and Medium Enterprises (MSMEs) and low carbon sustainable infrastructure plans will be implemented for rapidly growing urban areas. Low carbon mobility modes like electric vehicles, rail and DFCs are preferred endogenously because of share constraint in transport sector. More people opt to work from home. This is applied through exogenous transport demand reduction. Efficient cooking and greener technologies are adopted by the households mostly because of device improvement coefficients. Besides, ways to reduce, reuse and recycle materials is encouraged in all sectors.

[1] PAT scheme is part of National Mission on Enhanced Energy Efficiency (NMEEE) implemented by Bureau of Energy Efficiency (BEE) under Ministry of Power. PAT is a market-based mechanism where energy saving certificates can be traded by specified energy-intensive industries.

2 Degree High Growth (2DegHG) scenario assumes higher potential average annual labour productivity growth rate of 5.8% from 2013-2050 (Table 5.1). Also India will achieve national development targets aligned with SDGs like sustainable infrastructure for cities, increased share of manufacturing sector in GDP, universal electricity access, housing for all and adequate safe water for everyone in rural India by the year 2022. This will lead to increase in the demand for energy from industries like iron and steel, cement, chemical and petrochemical, textile, aluminium and residual industries along with demand for transport, construction and other services. On the other hand, development needs to be aligned to the low carbon pathways so as to limit the global temperature rise to 2°C above the pre-industrial levels. In this scenario carbon emissions remain at the same level as 2DegMG scenario, however the capital costs rise significantly to support 'green' high growth. Coal continues to remain the mainstay of Indian power sector up until 2030 It is a primary energy source for energy intensive industries considering the rapid growth in demand of steel, cement, aluminium and chemicals. Significant investment will be made in clean coal technologies like IGCC, CCS and BECCS. By 2050, older inefficient coal plants will be phased out, renewables will be integrated in the grid reducing the coal dependence, battery and pumped storage technologies will provide cost effective energy storage options for grid reliability. So the capital intensities in the energy sector increase and energy intensities in industrial and services sector decrease correspondingly. Petrol and diesel demand will diminish with electric vehicles dominating the transport sector and an increase of shared transport. Demand-side interventions like clean cooking options of electric cooking, LPG and natural gas will be provided to all thus eliminating the dependence on inefficient cooking fuels like firewood. Households will adopt clean technologies like LED for lighting, energy efficient appliances based on renewable energy for cooling and heating and public transport for commutation. Energy consumption in services sector is critical as well considering the fact that it is the fastest growing sector in India and currently constitutes largest share in GDP. Behavioural changes in services sector will play vital role like adoption of energy efficient appliances, use of solar energy for energy needs and provision of work from home option. Disruptive technologies will be used in energy-intensive sectors like switching to electric furnace in iron and steel sector and alternate materials in place of cement or steel in construction of buildings, roads, ports and other infrastructure. This suggests that energy-efficiency improvement is pivotal but with the caveat of increase in capital intensity in energy intensive productive sectors. AIM end-use trajectories demonstrate that the thrust lies on driving the low carbon pathways early on in this scenario before the technology or behavioural lock-ins build-up. Besides, the heat losses are significantly reduced and waste heat is recovered for reuse. Recycling, reuse and dematerialization techniques will be employed in all sectors.

In 2 Degree Low Growth (2DegLG) scenario, annual average labour productivity growth rate of 4.5% is assumed. The carbon emissions remain same as in 2DegMG scenario. The relaxed carbon constraint and reduced end-use demands leads to the choice of less efficient cheap technologies in AIM/Enduse. Lower capital costs feed back to IMACLIM and allow higher growth i.e. a smaller efficiency gap between potential growth (labour supply x labour productivity increases) and actual growth. The energy efficiencies of the technologies used in various industrial sectors is low and the capital input per unit output is less comparatively. There is limited capital investment in reducing energy consumption in sectors like iron and steel, cement, aluminium and other manufacturing industries compared to high growth scenario. Renewables contribute significantly to power generation

in long term but energy efficiency in electricity sector remains low. The costly technologies like CCUS, smart grid and IGCC will not be employed.

5.4 Results and discussion

We first explore the energy dynamics for our reference scenario (BAU), which emulates the current policy, socio-economic and energy prospects. We find that coal dominance persists from base year 2012 (BY2012) up to 2050, although the contributions of renewables,[1] biomass and natural gas increase as a result of current climate policies (Figure 5.2). This is expected as the demand for all goods in Indian economy, which is largely fossil-dependent, is growing rapidly. In 2050, power generation is the largest buyer of energy and coal dominates its mix. Other large energy-consuming sectors are residual industries, transport, iron & steel and other services. Coal, transport fuels (referred as TRANSPFUEL), non-transport refined petroleum fuels (referred as OILNTFUEL) and natural gas constitute major shares of energy consumption in BAU2050. As India is largely dependent on imports for meeting demand of fossil fuels such as crude oil, natural gas and coal, it must have macroeconomic repercussions.

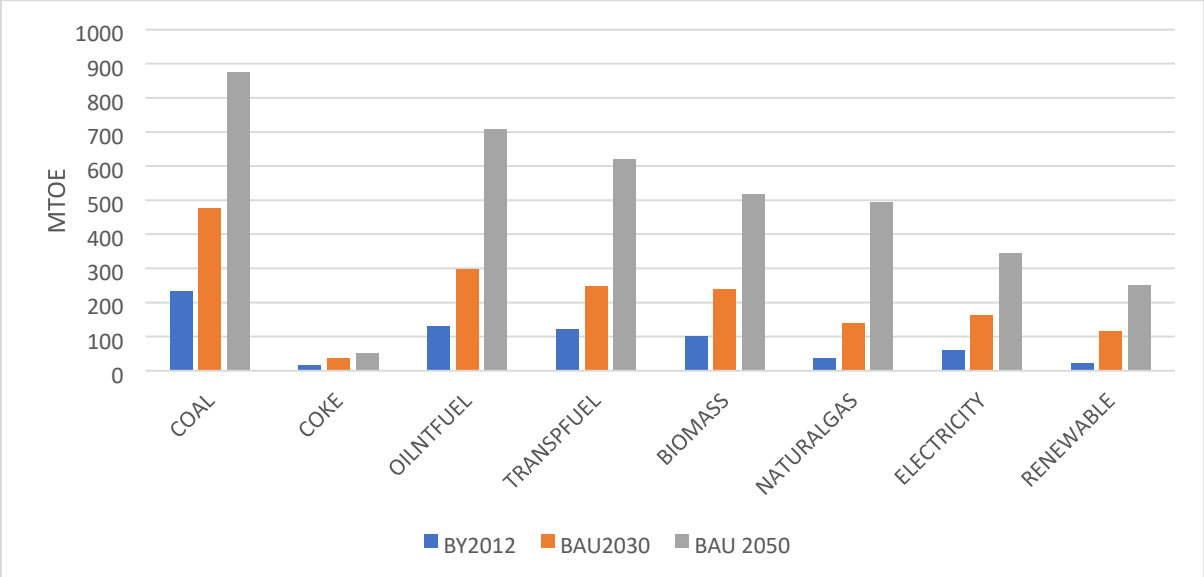

Figure 5.2 Energy supply in BAU scenario (MTOE)

Source: IMACLIM-IND and AIM/Enduse coupled simulations. The reported supply is the specific supply of commercial flows as tracked by the output volumes of CGE models. BY2012 stands for "base year 2012".

Our focus in the paper being macroeconomic assessment, we now turn to investigating the macroeconomic results of all scenarios for both horizon years (Table 5.2). As regards real GDP,

[1] To facilitate the forcing of AIM/Enduse investment trajectories, we single out renewables into one production sector, whose output's exclusive use is as input to the electricity sector.

scenarios expectedly perform according to their potential growth assumptions, i.e. the lower growth scenario less than the BAU and middle growth scenarios, and the latter scenarios less than the higher growth scenario.

At 2030, the 2-degree scenarios perform 0.1 percentage point below their potentials (compare Table 5.1 to Table 5.2), while the BAU is 0.2 points below potential. Comparison between 2DegMG and BAU, who share the same labour productivity assumptions, reveals that the low-carbon constraint is compatible with maintained and indeed slightly increased economic activity, despite its higher investment costs in energy supply and demand (see below), which translate into a higher ratio of energy expenses to GDP. This is at least under our assumptions of public policies controlling the investment effort (Johansen closure) and adjusting the exchange rate to warrant real wages increasing as labour productivity (our trade specification). As we indicated in Section 5.2, these specifications require qualifying activity results with attention to domestic and foreign savings evolutions.

Table 5.2 Macroeconomic results of all scenarios

	2012	**2030**				**2050**			
	BY2012	BAU	2DegMG	2DegHG	2DegLG	BAU	2DegMG	2DegHG	2DegLG
Real GDP (Trillion US dollar 2012)	1.7	5.2	5.3	6.3	4.5	14.4	14.6	19.9	10.5
Average annual real GDP growth (%) (2013 onwards)	-	6.3%	6.4%	7.4%	5.4%	5.7%	5.8%	6.6%	4.8%
Trade balance ratio to GDP (%)	-7.4%	-9.6%	-8.8%	-8.0%	-10.0%	-10.6%	-9.4%	-7.8%	-12.7%
Household consumption ratio to GDP (%)	60%	68.1%	67.3%	66.5%	68.5%	69.0%	67.9%	66.3%	71.1%
Foreign debt ratio to GDP (%)	-128%	-149%	-133%	-113%	-160%	-218%	-183%	-132%	-286%
E imports ratio to GDP	10.0%	9.2%	8.2%	7.1%	10.0%	8.4%	7.3%	4.4%	11.4%
E expenses ratio to GDP	22.3%	24.5%	25.2%	23.4%	26.1%	28.2%	26.9%	26.5%	31.0%
Emission intensity (ktCO2eq per 1000 US$ 2012)	1.1	0.8	0.6	0.5	0.7	0.4	0.2	0.1	0.2

Source: IMACLIM-IND and AIM/Enduse coupled simulations.

The impact of scenarios on foreign savings follows on their trade impacts. The 2-degree scenarios reduce the weight of energy imports only in their medium and high growth variants, whereas low growth increases it. Under our assumption of endogenous trade, reduced energy imports translate into improved trade balances compared to BAU, although terms-of-trade appreciation mitigates the improvement. Because of our choice of trade specification, the improved trade balance does not affect activity but reduces the cumulated foreign debt. However, it also constitutes an effective decrease of foreign savings. This decrease prompts a compensating increase of domestic savings, which in turn decreases household consumption.

Comparison of the BAU and 2DegMG scenarios summarises the revealed macroeconomic tradeoffs. By 2050 and under the assumption of appropriate investment and exchange rate policies, the lowcarbon pathway has the potential to reduce the foreign debt significantly (a 35-GDP point decrease) via improvement of the trade balance, at the consumption cost of 1% of GDP, while slightly improving activity. In the event of high growth, the low-carbon option achieves close to maintaining foreign debt at its base-year level (+4 GDP points only by 2050) by further decreasing the share of household consumption in GDP by 1.6 points—but for a largely improved GDP. Conversely, the BAU trajectory and even more so the low-growth low-carbon trajectory lead to dramatic increases of the foreign debt (respectively +90 and +158 GDP points by 2050) via increased trade deficits.

Turning to activity composition, the 2030 GDP shares of industries, agriculture and services decrease in the 2DegMG scenario compared to BAU (Figure 5.3). The reason is that the share of valueadded mobilized by energy sectors is higher in the 2DegMG scenario because of the higher capital costs of the technologies required to achieve advanced mitigation targets: super-critical and ultra-super-critical coal based power plants, Carbon Capture and Storage (CCS) options, Integrated Gasification Combined Cycle (IGCC) and Pressurized Fluidised Bed Combustion (PFBC). By 2050, however, the GDP share of industries also increases, under the combined pressure of the increased capital intensity induced by penetration of energy-efficient techniques (as depicted by AIM) and the increased rental price of capital induced by the consecutive pressure on capital demand—which affects capital-intensive industrial activities more than other activities. The higher investment costs of meeting the same mitigation objectives in the higher growth scenario show both in the increased share of energy sectors (+1.9 GDP points in 2050) and industries (+2 GDP points in 2050).

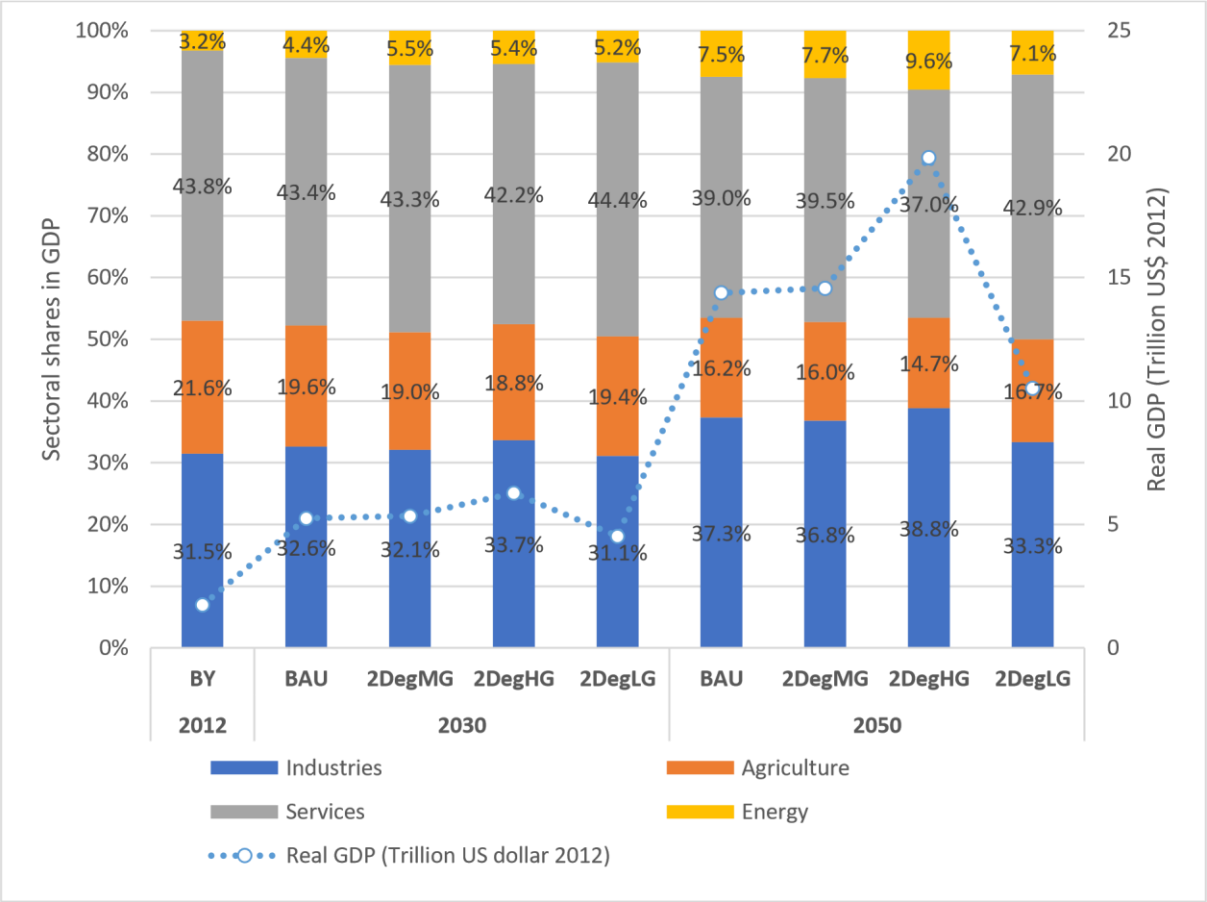

Figure 5.3 Total GDP and sectoral breakdown at base year (BY) 2012, in 2030 and 2050.

Source: IMACLIM-IND and AIM/Enduse coupled simulations.

Past data on infrastructure investment in India show that more than 40% of the total investments have been going into the energy supply sector (OECD, 2014). Nearly 77 billion USD have been invested in energy supply annually on average since 2010 (IEA, 2015). However, this capital infusion might not be sufficient to achieve low-carbon scenarios in the future. By 2030, the gap with respect to BAU investment is 12.6 Billion USD in the 2DegMG scenario, while it increases to 26.1 billion USD in the 2DegHG scenario (Table 5.3). Assuming constant growth rates from 2012 on, this amounts to a cumulative increase of 300 billion USD in energy supply investment from 2015 to 2040 in the 2DegMG scenario compared to BAU. This is a significant amount compared to the estimated cumulative investment in
Indian energy supply of around 2 trillion USD from 2015 to 2040 in the New Policies Scenario of the International Energy Agency, which mainly considers the NDCs submitted to UNFCCC (IEA, 2015). In the 2DegHG scenario, capping carbon emissions at the same level as in the 2DegMG scenario induces rising capital costs to support 'green' high growth.

Table 5.3 Gap to BAU of 2030 investments in energy supply for low-carbon scenarios

	Energy supply investment* (Billion 2012 USD)	Gap to BAU (Billion 2012 USD)
2DegMG	223.7	12.6
2DegHG	237.2	26.1
2DegLG	159.9	-51.1

*Projections of energy supply investments for 2030 from IMACLIM-IND model
Source: IMACLIM-IND and AIM/Enduse coupled simulations.

Further sectoral detail reveals that the increase of energy supply investment under tighter carbon constraint (comparing 2DegMG to BAU) happens both in the power sector and in non-fossil energy supply, i.e. renewables and biomass supply (Figure 5.4). Beyond energy sectors, the transport and housing sectors also register significant investment increases prompted by the development of public transports and the relatively high income-elasticity of housing services consumption. The substantially higher growth of the 2DegHG scenario induces increased investment expenses in all sectors. Particularly, non-fossil energy supply mobilises largely higher resources to allow meeting the carbon constraint under increased activity. Conversely, the 2DegLG scenario has investment expenses lower than the BAU except, interestingly, for the power sector, whose capital efficiency appears decreased.

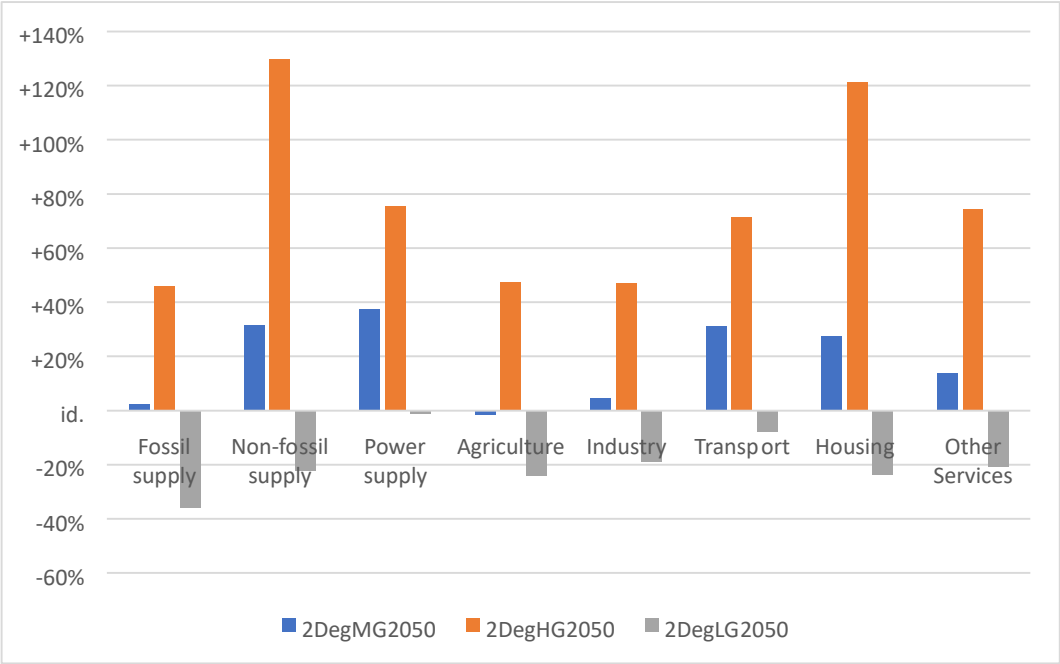

Figure 5.4 Variations of 2050 sectoral investments for three scenarios compared to BAU.

Source: IMACLIM-IND and AIM/Enduse coupled simulations.

Compared to BAU, the higher carbon constraint enforced on 2°C scenarios results in decreased (commercial) energy intensity and increased capital intensity of aggregate energy supply (Table 5.4).

These are endogenous results of AIM/Enduse under our climate policy assumptions, forced into IMACLIM-IND with consistency on all considered feedbacks after iteration to convergence. In the highgrowth scenario, the capital intensity of aggregate energy supply diverges by +0.6% on average annually from 2012 to 2050, while its energy intensity diverges by -0.3%, compared to BAU. This affects real GDP to some extent, as testifies the larger gap between realised growth (at 6.6% annual average) and potential growth (at 7.0% annual average).

Table 5.4 Deviations of energy and capital intensity of energy supply from BAU to 2°C scenarios

	2DegMG	2DegHG	2DegLG
Energy intensity variation from BAU (on average annually from 2012 to 2050) *	-0.11%	-0.34%	-0.04%
Capital intensity variation from BAU (on average annually from 2012 to 2050) *	+0.12%	+0.57%	+0.03%

*On average for all energy supply sectors of IMACLIM-IND, not accounting for solar and wind inputs to power.
Source: IMACLIM-IND and AIM/Enduse coupled simulations.

In the 2DegHG scenario, coal remains the mainstay of the Indian power sector up until 2030. It is also the main energy source for energy-intensive industries. Consequently, significant investment goes into clean coal technologies. By 2050, older inefficient coal plants have phased out, renewables have developed and battery and pumped storage technologies provide cost-effective energy-storage options to guarantee grid reliability. The aggregate capital intensities of energy supply increases and the aggregate energy-intensity of energy supply decreases correspondingly (Table 5.4). In the 2DegLG scenario, the energy efficiencies of the technologies used in various industrial sectors are low and the capital input per unit output is less, comparatively. The contribution of renewables to power generation does increase in the long term but the energy efficiency of the electricity sector remains low.

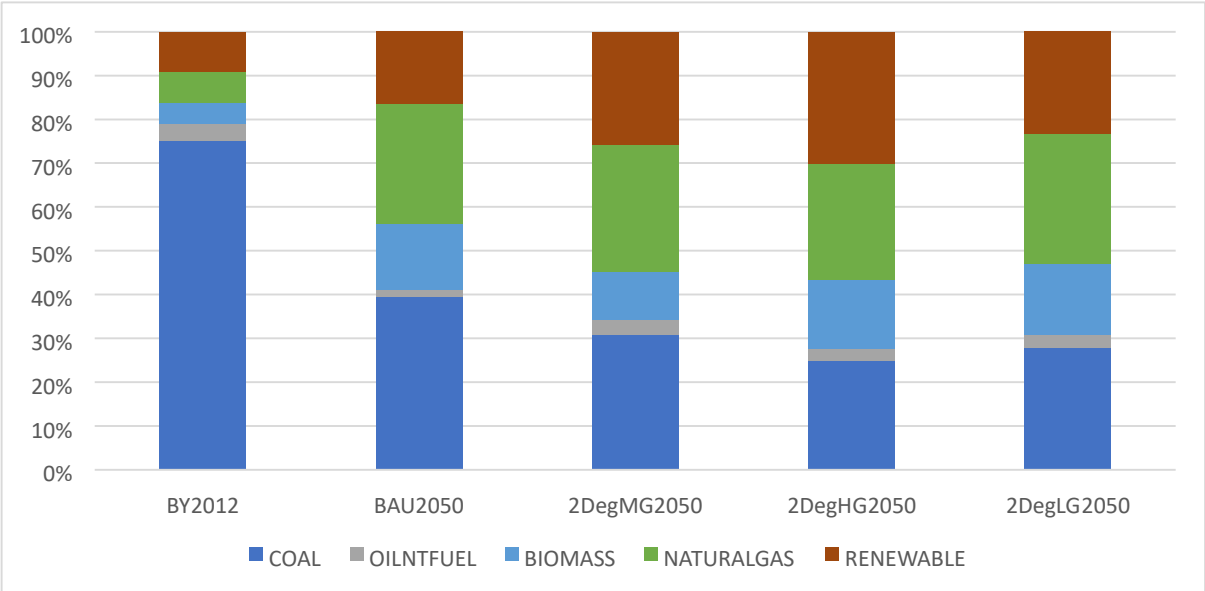

Figure 5.5 Power generation by source in 2012 and for 4 scenarios in 2050. Source: IMACLIM-IND and AIM/Enduse coupled simulations.

All scenarios enforce carbon constraints that induce shifts of the energy mixes and consumption choices of all agents. One key source of emissions is the power sector. The 165Bt carbon constraint in BAU allows maintained coal dominance. The more stringent 123Bt constraint of 2°C scenarios translates into renewables and natural gas gaining shares (Figure 5.5). Another important source of emissions is households' energy consumptions. The constraints translate in a drastic decline of transport fuels consumptions in the medium growth scenario, mainly due to the shift to electric vehicles and public transport (Figure 5.6). Non-transport petroleum fuels consumptions decline as well due to the massive substitution of electric stoves and solar cookers to kerosene and firewood stoves. Under the 2DegHG scenario, all households' energy consumptions increase compare to 2DegMG because of substantially higher income. However, the consumptions of fossil fuels remain below BAU levels (compare Figures
5.6 and 5.7).

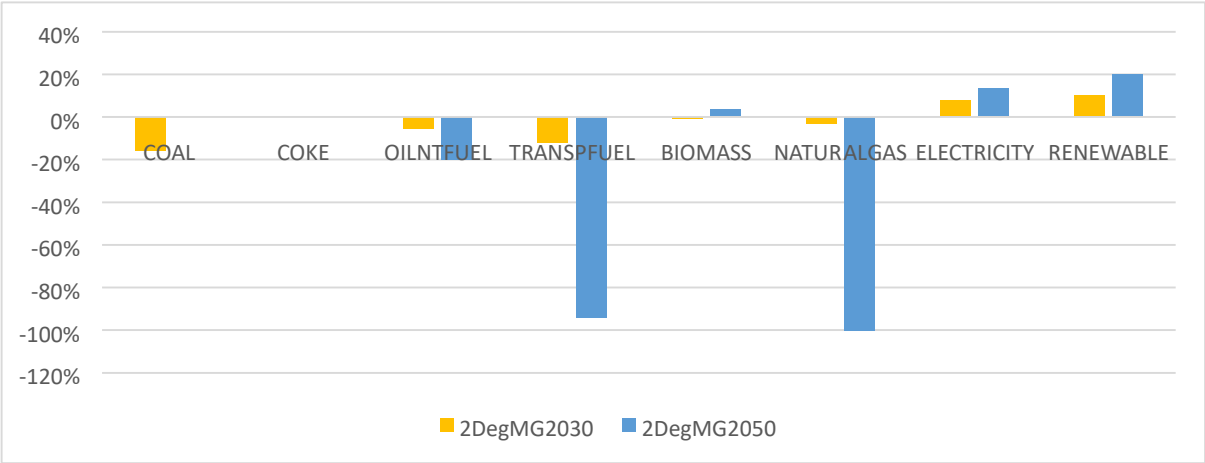

Figure 5.6 Household consumption change (%) in 2DegMG scenario compared to BAU. Source: IMACLIM-IND and AIM/Enduse coupled simulations. Coal household consumption in both BAU and 2DegMG in 2050 is nil. Hence, there is no corresponding bar in figure. Biomass consumption change in 2DegMG2030 compared to BAU2030 is almost nil.

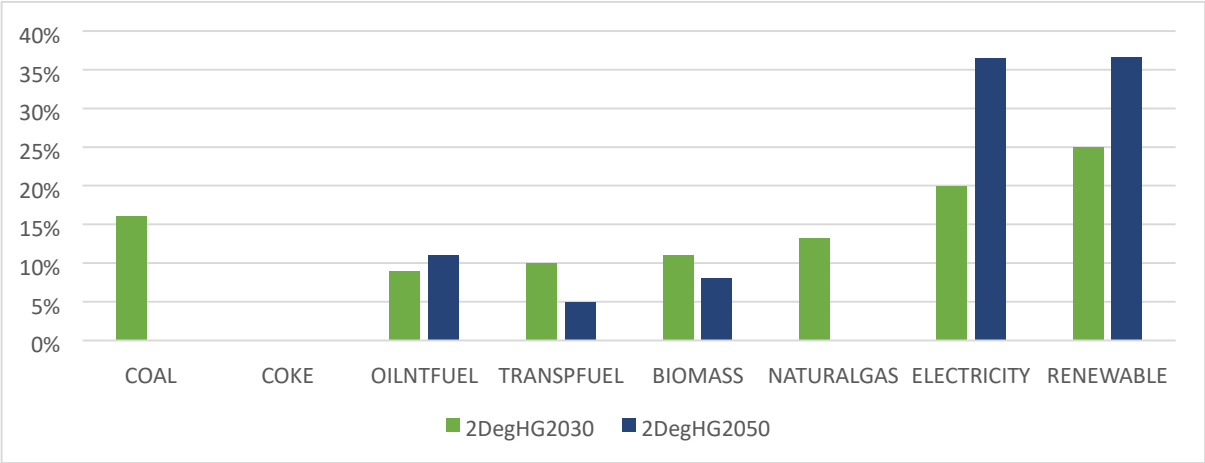

Figure 5.7 Household consumption change (%) in 2DegHG scenario compared to 2DegMG.

Source: IMACLIM-IND and AIM/Enduse coupled simulations.

Overall, the 2DegHG scenario stands out as not only the one with higher activity—somewhat by construction, but also because the costs of capping emissions do not weigh enough on national income to overly increase the gap to potential growth—but also the one with the lower trade balance deficit, which allows it to keep the Indian foreign debt under control. The main trade-off for these results is the lower share of GDP devoted to household consumption because of lower foreign savings. Another trade-off, hardly perceptible at our level of aggregation, is the potential employment loss and

financial and strategic loss of certain stakeholders in carbon-intensive activities or technologies during the process of low-carbon transition. Measures that could allow a smooth transition include retraining and job creation for workers in carbon intensive sectors, similar to what Germany provided to its ex-coal miners; "just transitions" where all stakeholders are affected equitably; "golden handshakes" consisting in single lump-sum payment to harmed employees such as Poland gave to workers in retreating sectors including again coal mining (Caldecott et al., 2017).

Our results additionally show that the capital intensity of energy supply increases significantly in the 2DegHG scenario to support clean growth and higher energy efficiency. This puts pressure on the rental price of capital considering limited supply. More pressure on investment markets comes from making energy-efficient the fast growing sectors of iron & steel and cement—to the point that the GDP share of industries increases in the long term in our 2°C scenarios compared to BAU. High growth can thus be clean but this requires more investment in non-fossil power generation and technology upgrades in energy-intensive industries like iron & steel, cement, aluminium and textile. In the event of higher growth, the services sector becomes critical because of its increasing contribution to GDP and hence to energy consumption. Behavioural changes in both the services sector and households' actions will be necessary to achieve the low carbon targets with high growth.

We acknowledge the fact that our modelling framework has a fixed technology portfolio and that

there is a chance that actual technical progress by 2050 eludes its grasp. There is for instance the possibility of transformation of fundamental end-uses such as graphene replacing cement; there is also the possibility of some technological breakthrough allowing transition to a fully non-fossil economy. The probabilities of such extreme transformations are low enough for our projected futures to provide valuable insights about achieving economic development along low-carbon pathways.

5.5 Conclusion and policy implications

This chapter uses the coupling of the IMACLIM-IND economic model and the AIM/Enduse energy systems model to determine the implications of low carbon pathways on Indian development. The analysis above can be used to draw the following conclusion and policy implications. First, low carbon policy measures like stringent energy efficiency requirements, promotion of electric vehicles and shared transport, switch to non-fossil fuel based power generation and giving incentives for behavioural adjustment are compatible with high growth targets. This will lead to improvement in trade balance thus providing the co-benefit of managing the energy security issue. Therefore, the policy makers need to decide the investments judiciously so as to achieve the twin goals of higher growth and low carbon economy concurrently. However, the switch to green technologies and non-fossil energy sectors in the fossil based economy like India can possibly lead to various socio-economic issues. Therefore, policymakers need to reconcile the conflicting interests of entrenched businesses in retreating sectors like coal and oil and the emerging low-carbon technologies like renewables.

Second, a fast developing country like India with increasing population and rapid urbanization has huge investment needs. Moreover, large share of the investment will go into energy supply to support green growth. Future projections show that the gap in required investment for low carbon pathway and current trend is large. International financial assistance will be critical since India has

huge development needs to focus on compared to the developed countries. Third, AIM has a fixed technology portfolio and there is a chance that the actual technical progress by 2050 is quite different. For instance, the possibility of changing the fundamental end uses in the economy like moving to graphene in place of cement or having a completely non-fossil fuel based economy. It is critical that lock-in of capital in fossil fuel based technologies is avoided to achieve a low carbon high growth economy. This will entail continuous evaluation of climate policies.

Lastly, as India experiences structural change in future, large number of Small and Medium Enterprises (SMEs) is expected to rise especially with large share of population engaged in agriculture shifting to industries and services sector as also shown in the results for 2050. These enterprises need to employ necessary capital in sustainable practices and techniques. In conclusion, the objective of the policymakers should be development of all economic sectors such that it is sustainable for achieving low carbon high growth economy. The trade-off here is that high growth scenario requires finances and technologies eventually benefitting from labour productivity gains and economies of scale.

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